

Number System

* Binary	Octal	Decimal	Hexadecimal
Radix = 2	Radix = 8	Radix = 10	Radix = 16
Unique digit :- 0, 1	Unique digit :- 0, 1, 2, 3, 4, 5, 6, 7	Unique digit :- 0, 1, 2, 3, 4, 5, 6, 7, 8, 9	Unique digit :- 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
Sequence: 0 1 10 11 100 101 110 111 1000	Sequence: 0 10 ... 70 100 1 11 ... 71 101 2 12 ... 72 102 3 13 ... 73 103 4 14 ... 74 104 5 15 ... 75 105 6 16 ... 76 106 7 17 ... 77 107	Sequence: 0 10 ... 90 100 1 11 ... 91 101 2 3 4 5 6 7 8 9 19 ... 99 109	Sequence: 0 10 ... 90 100 1 11 ... 91 101 2 12 3 13 4 14 5 15 6 16 7 17 8 18 9 19 A 1A B 1B C 1C D 1D E 1E F 1F 9F AF FF

* Any Number System to Decimal Number System Conversion

1. $(10101.1101)_2$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \quad 2^{-1} \quad 2^{-2}$

Sol $\Rightarrow 21 + \frac{1}{2^4} = (21.0625)_{10}$

2. $(714.35)_8 = 7 \times 8^2 + 1 \times 8^1 + 4 \times 8^0 + 3 \times 8^{-1} + 5 \times 8^{-2}$

$= (460.45)_{10}$

3. $(A2.C37)_{16} = A \times 16^1 + 2 \times 16^0 + 12 \times 16^{-1} + 3 \times 16^{-2} + 7 \times 16^{-3}$

$= 160 + 2 + \frac{12}{16} + \frac{3}{256} + \frac{7}{4096}$

$= (162.763)_{16}$

4. $(45.12)_6 = 4 \times 6 + 5 + \frac{1}{6} + \frac{2}{36}$

$= 29 + \frac{1}{6} + \frac{2}{36} = (29.22)_6$

$$5) (25.31)_9$$

$$\text{Sol } 2 \times 9 + 5 + \frac{3}{9} + \frac{1}{81}$$

$$> 23 + \frac{27+1}{81}$$

$$> 23 + \frac{28}{81} = (23.34)_{10}$$

* Decimal number system to any number system-

$$1) (29.20)_{10} \longrightarrow ()_2$$

$$\text{Sol } \begin{array}{r|l} 2 & 29 \\ \hline 2 & 14 \\ 2 & 7 \\ 2 & 3 \\ 2 & 1 \\ & 0 \end{array} \quad \begin{array}{l} 1 \uparrow \\ 0 \\ 1 \\ 1 \\ 1 \end{array} \quad (29)_{10} = (11101)_2$$

$$0.20 \times 2 =$$

$$= 0.40 \times 2$$

$$= 0.80 \times 2$$

$$= 1.60$$

$$= 0.60 \times 2$$

$$= 1.20$$

$$\begin{array}{l} 0 \\ 0 \\ 1 \\ 1 \end{array} \downarrow$$

$$(0.20)_2 = (0011)_2$$

$$(29.20)_{10} \longrightarrow (11101.00110)_2$$

$$0.20 \times 2 = 0.40$$

$$0.40 \times 2 = 0.80$$

$$0.80 \times 2 = 1.60$$

$$0.60 \times 2 = 1.20$$

$$0.20 \times 2 = 0.40$$

② $(321.35)_{10} \longleftrightarrow ()_8$

Sol

8	321
8	40
8	5
	0

$\rightarrow 1$
 $\rightarrow 0$
 $\rightarrow 5$

$0.35 \times 8 = 2.80$

$0.80 \times 8 = 6.40$

$0.40 \times 8 = 3.20$

$0.20 \times 8 = 1.60$

$0.60 \times 8 = 4.80$

$0.80 \times 8 = 6.40$

$(0.35)_{10} = 263146$

$(321.35)_{10} \longleftrightarrow (501.263146)_8$

③ $(1250.25)_{10} \longleftrightarrow ()_{16}$

Sol

16	1250
16	78
16	4
	0

2
 14
 4

$(2E2)_{16}$

$0.25 \times 16 = 4.00$

$0.00 \times 16 = 0.00$

$(.4)_{16}$

$(1250.25)_{10} \longleftrightarrow (2E2.40)_{16}$

Que. $(523.4)_6 \longleftrightarrow ()_9$

$\searrow$
 $()_{10}$
 $\nearrow$

Sol $(523.4)_6$

$$\begin{array}{c} 523 \\ \downarrow \\ 5 \times 6^2 + 2 \times 6^1 + 3 \times 6^0 + \frac{4}{6} \end{array} \Rightarrow (195.66)_{10}$$

9	195	
9	21	→ 6
9	2	→ 3
	0	→ 2

$$0.66 \times 9 = 5.94$$

$$0.94 \times 9 = 8.46$$

$$0.46 \times 9 = 4.14$$

$$0.14 \times 9 = 1.26$$

$$0.26 \times 9 = 2.34$$

$$(523.4)_6 \longrightarrow (236.5841)_9$$

Que. The decimal equivalent of Binary number $(10110.11)_2$

Ans —

Sol $22 + \frac{2+1}{4} = (22.75)_{10}$

Que. $\sqrt{(224)_8} = (13)_8$ $r=9$

$$\Rightarrow \sqrt{2 \times r^2 + 2 \times r^1 + 4 \times r^0} = 1 \times r + 3$$

$$2r^2 + 2r + 4 = r^2 + 6r + 9$$

$$r^2 - 4r - 5 = 0$$

$$\frac{4 \pm \sqrt{16 + 20}}{2} \Rightarrow \frac{4 \pm 6}{2} = 5 \text{ or } -1$$

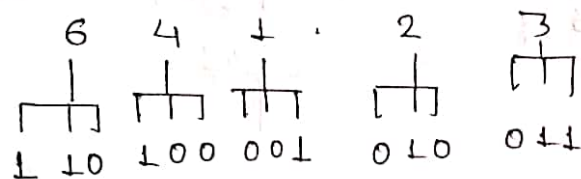
(r cannot be negative)

Que. The number of ones present in binary Representn of $15 \times 256 + 5 \times 16 + 3$ are 8.

Sol $15 \times 16^2 + 5 \times 16^1 + 3 \times 16^0$
 $(F53)_{16} \rightarrow 1111\ 101\ 011$

• Hexadecimal & Octal to Binary:

i) $(641.23)_8 \rightarrow (\)_2$



$\Rightarrow (110100001.010011)_2$

ii) $(C59.4)_{16} \rightarrow (\)_2$

$1100\ 0101\ 1001.0100$

$\Rightarrow (110001011001.0100)_2$

Que.

* Binary to Octal & Hexadecimal:

i) $(1101.10101)_2 \rightarrow (\)_8$

Sol $001\ 101.101010 \rightarrow (15.52)_8$

$1101.10101000 \rightarrow (D.A8)_{16}$

* Octal addition

$$\begin{array}{r} \overset{1}{7} \overset{1}{4} \overset{1}{2} \cdot \overset{1}{5} 3 \\ \underline{5 \ 4 \ 3 \cdot 2 \ 7} \\ (1 \ 5 \ 0 \ 6 \ 0 \ 2)_8 \end{array}$$

OR $10 - 8 = 2$

$$\begin{array}{r|l} 8 & 10 \\ 8 & \underline{1} \rightarrow 2 \uparrow \\ & 0 \rightarrow 1 \uparrow \end{array}$$

Base 6 addition:

$$\begin{array}{r} \overset{1}{3} \overset{1}{4} \overset{1}{3} \cdot \overset{1}{2} 5 \\ \underline{5 \ 3 \ 2 \cdot 3 \ 4} \\ (1 \ 3 \ 2 \ 0 \cdot 0 \ 3)_6 \end{array}$$

$$\begin{array}{r|l} 6 & 9 \\ 6 & \underline{1} \rightarrow 3 \\ & 3 \end{array}$$

don't divide
Subtract by
radix.

Que. $(135)_x + (144)_x = (323)_x$

Sol

$$\begin{array}{r} 1 \ 3 \ 5 \\ 1 \ 4 \ 4 \\ \hline \end{array}$$

note

~~9 - x = 3~~

$9 - x = 3$

$x = 6 //$

Que. $24 + 14 = 21$

$8 - x = 1$

$x = 7 //$

Que. $(2.3)_4 + (1.2)_4 = (7)_4$

Sol

$$\begin{array}{r} 3 \\ 2 \\ \hline 6 \leftarrow 4 = 2 \end{array}$$

$$\begin{array}{r} 2.3 \\ 1.2 \\ \hline 1 \ 0.1 \end{array}$$

$(10.1)_4 //$

Que $(235)_{R_1} = (565)_{R_2} = (865)_{R_2}$

Sol $2 \times R_1^2 + 3 \times R_1 + 5 =$

$$2R_1^2 + 3R_1 + 5 = 565$$

$$\cancel{-3 \pm \sqrt{9 - 40}}$$

$$\Rightarrow 2R_1^2 + 3R_1 - 560 = 0$$

$$\Rightarrow R_1 = \frac{-3 \pm \sqrt{9 + 560 \times 4 \times 2}}{4}$$

$$R_1 = 16$$

Que. $(43)_x = (y3)_8$ no. of possible solⁿ is

Sol $4x + 3 = 8y + 3$

$$4x = 8y$$

$$\frac{x}{y} = 2$$

$$x = 2y$$

$$\cancel{y = 1, 2, 3}$$

$$x = 8$$

	x	y
	2	1
	4	2
6		3
8		4
10		5
12		6
14		7
16		8

5 solⁿ

Que. $(123)_5 = (xy)_8$

Sol $25 + 10 + 3 = xy + 8$

$xy = 30$

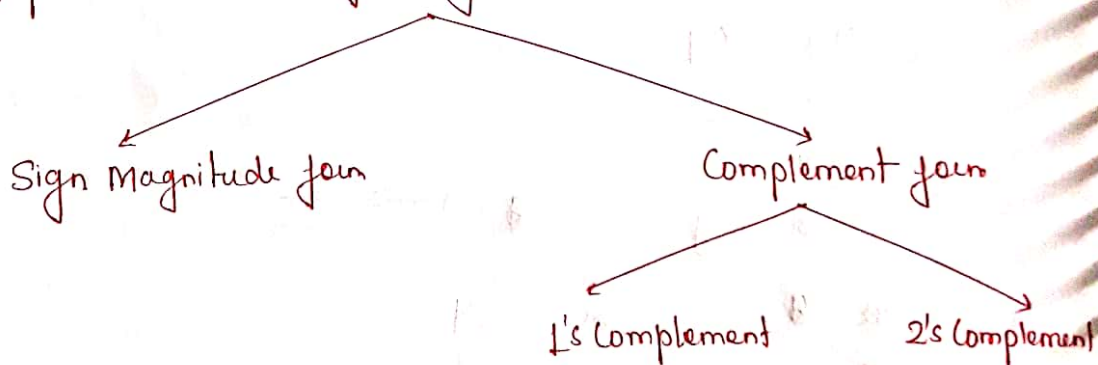
$y = \frac{30}{x}$

x 3 possible solⁿ

- ① $(x < y)$
② $y_{min} = 9$

x	y
1	30
2	15
3	10
4	7.5
5	6
6	5
30	1
15	2
10	3

* Representation of Signed binary Number



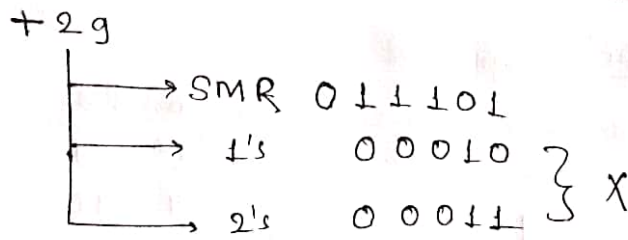
i) $-29 \rightarrow$ Signbit Magnitude (29)

$$\begin{array}{c} \underline{1 \ 11101} \\ \downarrow \quad \downarrow \\ \text{signbit} \quad \text{Magnitude bit} \end{array}$$

This is Sign Magnitude form.

ii) -29 in 1's Complement $\Rightarrow \underline{1 \ 00010}$
 Signbit 1's comp of 29

iii) -29 in 2's Complement $\Rightarrow \underline{1 \ 00010} + 1 = 1 \ 00011$



For positive number Sign Mag. = 1's Comp = 2's Comp

+29 =

Sign bit	SMR	0 1 1 1 0 1
1's		0 1 1 1 0 1
2's		0 1 1 1 0 1

Short cut of 2's Complement

1 0 1 1 0 1 1
 1's 2's 1's
 0 1 1 1 0 1
 as 1's

Que. find 2's Complement of 100011

Sol

1's Comp.	0 1 1 1 1 0 0
2's Com	0 1 1 1 1 0 0
	+ 1
	0 1 1 1 1 0 1 = -29 //

Note 2's Comp
 -ve no.
 1 0 0 0 1 1
 1's comp 2's comp
 as it is only this

Note: 2's Comp. of 2's Comp. of a number is number itself.
 1's Comp. of 1's Comp. of a no. is no. itself.

Que. 2's Comp. of Some numbers are given. Find out the original no.

i) 11010

Sol 101010 $\Rightarrow -6 //$

ii)

a) 0 0 0	→	1 1 1	} X numbers are positive
b) 0 0 1	→	1 1 1	
c) 0 1 0	→	1 1 0	
d) 0 1 1	→	1 0 1	
e) 1 0 0	→	0 1 1	} Special case
f) 1 0 1	→	1 0 1	
g) 1 1 0	→	0 1 0	
h) 1 1 1	→	1 0 0	} ✓
		1 0 0	

sign bit as it is

0 0 0	+0
0 0 1	+1
0 1 0	+2
0 1 1	+3

✓

Special Case of 2's Complement

1 0 0

no. of zeros
2

original no
 $-2^2 = -4$

1 0 0 0 0

4

$-2^4 = -16$

1 0 0 0 0 0

5

$-2^5 = -32$

special case in 1's complement form:

* The Range of decimal number that can be represented using n-bit 2's complement representation is

$$-(2^{n-1}) \text{ to } (2^{n-1}-1)$$

Que. 1's Complement representation of some no are given find out the original number.

Sol

a) 0 0 0	$\xrightarrow{1's}$	0 0 0	+ 0
b) 0 0 1		0 0 1	+ 1
c) 0 1 0		0 1 0	+ 2
d) 0 1 1		0 1 1	+ 3
e) 1 0 0		1 1 1	- 3
f) 1 0 1		1 1 0	- 2
g) 1 1 0		1 0 1	- 1
h) 1 1 1		1 0 0	- 0

Note:- In 1's Complement Representation we have two separate representation for +0 & -0 where as in 2's complement we have single representation for zero.

The Range of decimal numbers that can be represented using n-bit 1's comp. form is $-(2^{n-1}-1) \text{ to } +(2^{n-1}-1)$

Que. 11001, 1001 & 111001 Correspond to 2's Comp Representation of which of the following Set of no.

- a) 25, 9, 57 b) -6, -6 & -6 c) -7, 7 & 7
d) -25, -9, -57

Sol.

11001	→	101110	= -7
1001	→	1111	→ -7
111001	→	100111	→ -7

Que. +6 → 5 bit 2's Comp

0110 //

eg. +6 in 8 bit 2's Comp

min = 0110

0000 0110

in 7 bit 2's Comp 00000110

eg. -7 in 8 bit 2's

min. 1111

-7 1001

in 8 bit 1111 1001

note

Que. A no. in 4bit 2's Complement form is $x_3 x_2 x_1 x_0$. This no. when stored using 8 bit will be

Sol $x_3 x_3 x_3 x_3 x_3 x_2 x_1 x_0$

note

Que. Let $A = 11111010$
 $B = 00001010$ be two 8 bit 2's Comp. representⁿ. Their product in 2's Comp. is

Sol $11111010 \rightarrow 10000110 = -6$
 $00001010 \rightarrow 1010 = 10$

note
 There is no need of finding 2's c of the number $\rightarrow -60$

$\rightarrow 100111100$

$\rightarrow 10111100$
 2's

$11000100 //$

$\begin{array}{r} 32 \text{ --- } 1 \\ 16 \text{ --- } 1 \\ \hline 48 \text{ --- } 1 \\ 8 \text{ --- } 1 \\ \hline 56 \text{ --- } 1 \\ 4 \text{ --- } 0 \end{array}$

Que. Assuming all no. in 2's Complement which of the following no. is divisible by 11111011

Sol $11111011 \xrightarrow{2's} 10000101$
 $64+32+16+8+2+1 = 125$
 (-5)

A) $11100111 \xrightarrow{2's} -25$

B) 11100100

C) 1011101

D) 1111011

* Complements Arithmetic

1) 1's ~~True~~ Complement

eg. $-26 + 9$

let $-26 + 9 = x$

$$(\text{1's comp of } -26) + (\text{1's comp of } 9) = \text{1's comp of } x$$

$$\begin{array}{r} -26 \xrightarrow{1s} \begin{array}{cccccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{array} \end{array}$$

$$\begin{array}{r} 9 \xrightarrow{1s} \begin{array}{cccccc} 0 & 1 & 0 & 0 & 1 & \\ 0 & 0 & 1 & 1 & 0 & \end{array} \end{array}$$

Note
Answer is in 1's Comp.
form. So to get original
no. 1's comp. the answer

★
do not take compl.
of +ve no.

$$\begin{array}{r} \text{now } \begin{array}{cccccc} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{array} \\ \hline 1 & 0 & 1 & 1 & 0 & 1 \end{array} = 1011101$$

1's comp of 1011101 $\longrightarrow$ 1000010
-17 //

eg. $+26 - 9$

So $26 \longrightarrow 01101010$

$-9 \longrightarrow 1101110$

zeros used for +ve

Note
To adjust
equal bits
in both
numbers
for -ve numbers
1 is used

$$\begin{array}{r} \begin{array}{cccccc} 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \end{array} \\ \hline \begin{array}{cccccc} 1 & 0 & 1 & 0 & 0 & 0 \end{array} \\ \hline \begin{array}{cccccc} 0 & 1 & 0 & 0 & 0 & 1 \end{array} \end{array}$$

Note
Positive no
no need
to complement

Difference betn 1's & 2's Comp arithmetic

- In 1's Complement arithmetic we add the Carry (if generated) to the Result where as in 2's Complement arithmetic we ignore the Carry (if generated)

Q1.

Concept of Overflow

Q. Using 2's Comp arithmetic, Solve

- (1) $-2+3$ (2) $-3+2$ (3) $3+2$ (4) $-3-2$

Sol 1 $-2 \xrightarrow{2's} 110$
 $3 \xrightarrow{2's} 011$

$$\begin{array}{r} 110 \\ + 011 \\ \hline 1001 \end{array}$$

 Carry ignore $\Rightarrow 001$

let,

$$-2+3 = x$$

$$2^1C(-2) + 2^1C(3) = 2^1C(x)$$

$$001 \xrightarrow{2's} 001 \text{ (true)}$$

$$\begin{array}{r} -3 \xrightarrow{2's} 101 \\ 2 \xrightarrow{2's} 010 \end{array}$$

(II) $-3+2 =$

$$\begin{array}{r} 101 \\ + 010 \\ \hline 111 \end{array} \xrightarrow{2's} 101 = -1$$

(III) $3+2 =$

not possible

$$\begin{array}{r} +ve \ 011 \\ +ve \ 010 \\ -ve \ 101 \\ \hline 111 \end{array} \xrightarrow{2's} 111 = -3$$

(IV) $-3-2$

NP

$$\begin{array}{r} -ve \ 101 \\ -ve \ 110 \\ +ve \ 011 \\ \hline 011 \end{array} \xrightarrow{2's} 011 + 3$$

Note:

i) Overflow can never occur on addition of two numbers with opposite sign. Overflow may occur on addition of two numbers with same sign.

ii) Addition of two positive numbers should never give a negative no. If such condition occurs, it indicates overflow.

* iii) In MSB, if $C_{in} \oplus C_{out} = 1$ indicates overflow.

$$\begin{array}{r} C_{in} \ 0 \\ \downarrow \ 0 \ 1 \\ \ 0 \ 1 \ 0 \\ \hline C_{out} \ 0 \ 1 \ 1 \ 1 \end{array}$$

$$C_{in} \oplus C_{out} = 0 \text{ no overflow}$$

$$\begin{array}{r} C_{in} \ 0 \\ \downarrow \ 1 \ 0 \ 1 \\ \ 1 \ 1 \ 0 \\ \hline C_{out} \ 1 \ 0 \ 1 \ 1 \end{array}$$

$$C_{in} \oplus C_{out} = 1 \text{ overflow}$$

Que. (i)
$$\begin{array}{r} 1 \ 1 \ 0 \ 0 \\ 1 \ 0 \ 1 \ 1 \\ \hline 1) \ 0 \ 1 \ 1 \ 1 \\ 0 \oplus 1 = 1 \text{ OV} \end{array}$$

(ii)
$$\begin{array}{r} 1 \ 0 \ 0 \ 1 \\ 1 \ 1 \ 1 \ 0 \\ \hline 1) \ 0 \ 1 \ 1 \ 1 \\ 1 \oplus 0 = 1 \text{ OV} \end{array}$$

(iii)
$$\begin{array}{r} 0 \ 1 \ 1 \ 0 \\ 1 \ 0 \ 1 \ 0 \\ \hline 1) \ 0 \ 0 \ 0 \ 0 \end{array}$$
 { no need of checking opposite sign never gen. ov }

$$C_{in} \oplus C_{out} = 0 \text{ no overflow}$$

(iv)
$$\begin{array}{r} 0 \ 1 \ 1 \ 1 \\ 0 \ 1 \ 1 \ 0 \\ \hline 0) \ 1 \ 0 \ 0 \ 1 \\ 1 \oplus 0 = 0 \text{ OV} \end{array}$$

Que. Two 2's Complement no. having Sign bit x and y are added and the sign bit of the Result is z . Which boolean function indicates the occurrence of overflow.

$$\begin{array}{r} x \text{ ---} \\ y \text{ ---} \\ \hline z \text{ ---} \end{array} \quad \text{for overflow}$$

	x	y	z	overflow 'f'
	0	0	0	0
Sum of two 0's	0	0	1	1
	0	1	0	0
	0	1	1	0
Opposite Sign	1	0	0	0
	1	0	1	1
	1	1	1	0

x	y	z	
0	1	0	0
0	0	0	1

$$f = \bar{x}\bar{y}z + x y \bar{z}$$

Sol

x	$y+z$	$y+\bar{z}$	$\bar{y}+\bar{z}$	$\bar{y}+z$
0	0	1	0	0
0	0	0	0	1

$$(y+z) * (\bar{x}+\bar{z}) * (x+\bar{y})$$

$$> (y+z)(\bar{x}+\bar{z})(x+\bar{y})$$

Code Converter

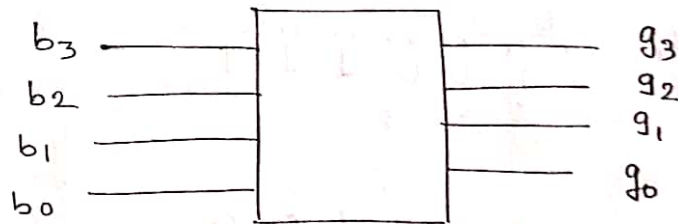
1. Binary to Gray (4bit)

	Input			
	b_3	b_2	b_1	b_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

	Output			
	g_3	g_2	g_1	g_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	1
3	0	0	1	0
4	0	1	1	0
5	0	1	1	1
6	0	1	0	1
7	0	1	0	0
8	1	1	0	0
9	1	1	0	1
10	1	1	1	1
11	1	1	1	0
12	1	0	1	0
13	1	0	1	1
14	1	0	0	1
15	1	0	0	0

Note
Gray Code

0	1	3	2
4	5	7	6
12	13	15	14
8	9	11	10



$$g_3(b_3, b_2, b_1, b_0) = b_3$$

g_2

	$b_1 b_0$	$\bar{b}_1 b_0$	$b_1 \bar{b}_0$	$\bar{b}_1 \bar{b}_0$
$b_3 \bar{b}_2$	0	1	3	2
$b_3 b_2$	4	5	7	6
$\bar{b}_3 \bar{b}_2$	12	13	15	14
$\bar{b}_3 b_2$	8	9	11	10

$$g_2 = \bar{b}_3 b_2 + b_3 \bar{b}_2$$

$$= b_2 \oplus b_3$$

g_1

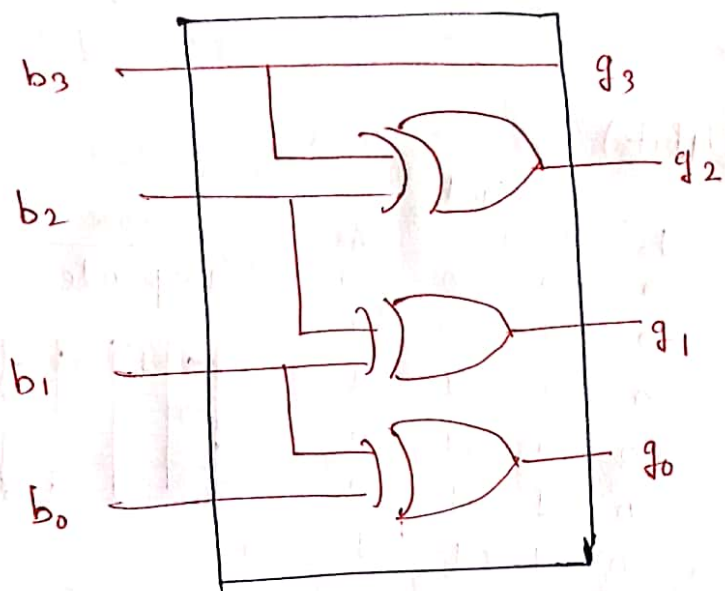
		1	1
1	1	0	0
1	1		
		1	1

$$g_1 = b_2 \oplus b_1$$

g_0

1			1
1			1
1			1
1			1

$$g_0 = b_1 \oplus b_0$$



ckt. Binary to Gray Converter

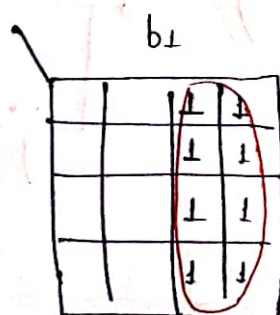
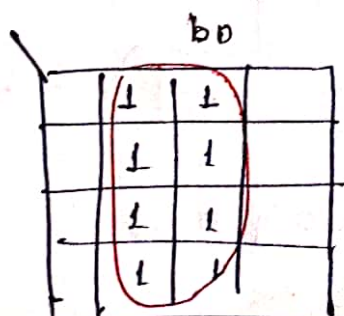
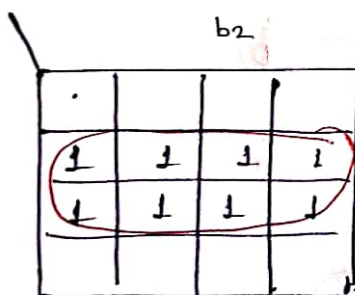
eg. $(0110)_{b_3 b_2 b_1 b_0} \longrightarrow \begin{matrix} g_3 = 0 \\ g_2 = 1 \\ g_1 = 0 \\ g_0 = 1 \end{matrix} \quad (0101)$

eg. $(0110)_2$
 $(0101)_{\text{gray}}$

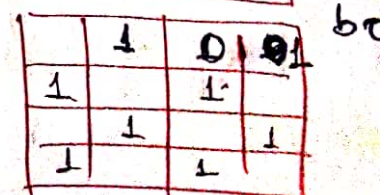
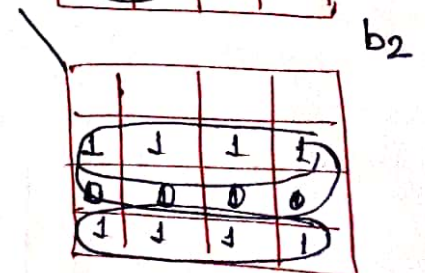
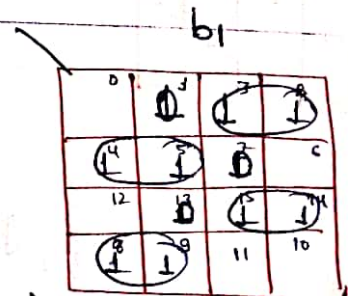
eg. 01101001
 01011010

* Gray to binary (4bit) Converter:

	g_3	g_2	g_1	g_0		b_3	b_2	b_1	b_0
0	0	0	0	0		0	0	0	0
1	0	0	0	1		0	0	0	1
3	0	0	1	1		0	0	1	0
2	0	0	1	0		0	0	1	1
6	0	1	1	0		0	1	0	0
7	0	1	1	1		0	1	0	1
5	0	1	0	1		0	1	1	1
4	0	1	0	0		1	0	0	0
12	1	1	0	0		1	0	0	1
13	1	1	0	1		1	0	1	0
15	1	1	1	1		1	0	1	1
14	1	1	1	0		1	1	0	0
10	1	0	1	0		1	1	0	1
11	1	0	1	1		1	1	1	0
9	1	0	0	1		1	1	1	1
8	1	0	0	0		1	1	1	1



~~b_2~~



$$b_3 = g_3$$

$$b_2 = g_3 \oplus g_2 = b_3 \oplus g_2$$

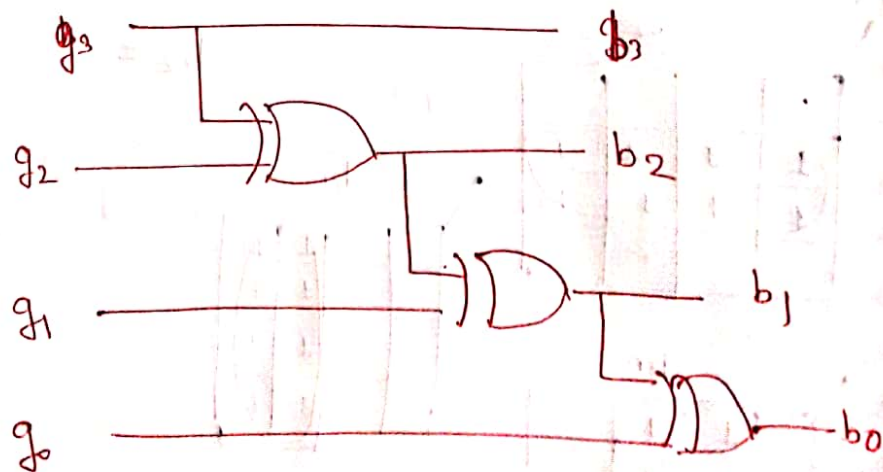
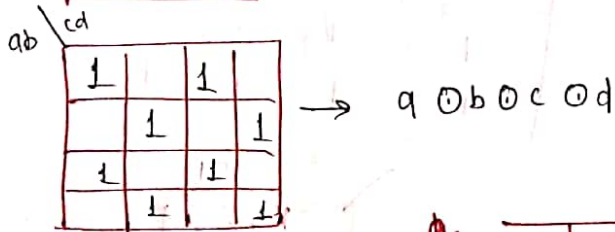
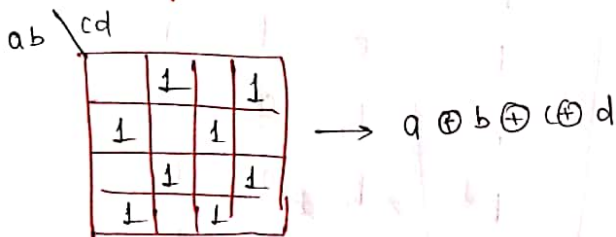
$$b_1 = g_3 \oplus g_2 \oplus g_1 = b_2 \oplus g_1$$

$$b_0 = \overline{g_3} \overline{g_2} (g_1 \oplus g_0) + \overline{g_3} g_2 (\overline{g_1} \oplus g_0) + g_3 \overline{g_2} (g_1 \oplus \overline{g_0}) + g_3 g_2 (\overline{g_1} \oplus \overline{g_0})$$

$$b_0 = (g_1 \oplus g_0) (\overline{g_3} \oplus g_2) + (\overline{g_1} \oplus g_0) (g_3 \oplus g_2)$$

$$b_0 = g_0 \oplus g_1 \oplus g_2 \oplus g_3 = b_1 \oplus g_0$$

Standard result



* delay of G to B Conv. is $\gamma \Delta t$ > delay of B to G Conv. Δt_{see}

eg. $\begin{array}{cccc} 1 & 1 & 0 & 0 \\ \downarrow \oplus & \downarrow \oplus & \downarrow \oplus & \downarrow \\ 1 & 0 & 0 & 0 \end{array}$ $\begin{array}{cccc} 1 & 0 & 0 & 1 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 1 & 1 & 0 \end{array}$

eg. $\begin{array}{ccccccccc} 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 \\ \downarrow & & & & & & & & \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{array}$

Que. $(24)_8 \rightarrow \text{gray}$
 $010100 \rightarrow 011110$

* Xs-3 Code to Bcs Code

x_3	x_2	x_1	x_0	b_3	b_2	b_1	b_0
0	0	1	1	0	0	0	0
0	0	1	0	0	0	0	1
0	1	0	1	0	0	1	0
0	1	1	0	0	0	1	1
0	1	1	1	0	1	0	0
1	0	0	0	0	1	1	0
1	0	0	1	0	1	1	1
1	0	1	0	1	0	0	0
1	0	1	1	1	0	0	1
1	1	0	0				

eg. $\begin{array}{cccc} 1 & 1 & 0 & 0 \\ \downarrow \oplus & \downarrow \oplus & \downarrow \oplus & \downarrow \\ 1 & 0 & 0 & 0 \end{array} \quad \begin{array}{cccc} 1 & 0 & 0 & 1 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 1 & 1 & 0 \end{array}$

eg. $\begin{array}{cccccccc} 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 \\ \downarrow & & & & & & & & \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{array}$

Que. $(24)_8 \rightarrow \text{gray}$
 $010100 \rightarrow 011110$

* Xs-3 Code to BCD Code

x_3	x_2	x_1	x_0	b_3	b_2	b_1	b_0
0	0	1	1	0	0	0	0
0	0	1	0	0	0	0	1
0	1	0	1	0	0	1	0
0	1	1	0	0	0	1	1
0	1	1	1	0	1	0	0
1	0	0	0	0	1	1	0
1	0	0	1	0	1	1	1
1	0	1	0	1	0	0	0
1	0	1	1	1	0	0	1
1	1	0	0				

B_0

$x_3 \backslash x_2 \backslash x_1 \backslash x_0$	$\bar{x}_3 \bar{x}_2 \bar{x}_1 \bar{x}_0$	$\bar{x}_3 \bar{x}_2 x_1 \bar{x}_0$	$\bar{x}_3 x_2 \bar{x}_1 \bar{x}_0$	$\bar{x}_3 x_2 x_1 \bar{x}_0$
$x_3 \bar{x}_2$	X	X		X
$x_3 x_2$	1		1	1
$\bar{x}_3 x_2$	1	X	X	X
$x_3 \bar{x}_2$	1			1

$$B_0 = x_3 x_1 x_0 + \bar{x}_3 x_3 + \bar{x}_1 \bar{x}_0$$

$$B_0 = \bar{x}_1 \bar{x}_0 + x_1 \bar{x}_0$$

B_1

$x_3 \backslash x_2 \backslash x_1 \backslash x_0$	$\bar{x}_3 \bar{x}_2 \bar{x}_1 \bar{x}_0$	$\bar{x}_3 \bar{x}_2 x_1 \bar{x}_0$	$\bar{x}_3 x_2 \bar{x}_1 \bar{x}_0$	$\bar{x}_3 x_2 x_1 \bar{x}_0$
$\bar{x}_3 \bar{x}_2$	X	X		X
$\bar{x}_3 x_2$		1		1
$x_3 \bar{x}_2$		X	X	X
$x_3 x_2$		1		1

$$B_1 = \bar{x}_1 x_0 + x_1 \bar{x}_0$$

$$= x_0 \oplus x_1$$

B_2

$x_3 \backslash x_2 \backslash x_1 \backslash x_0$	$\bar{x}_3 \bar{x}_2 \bar{x}_1 \bar{x}_0$	$\bar{x}_3 \bar{x}_2 x_1 \bar{x}_0$	$\bar{x}_3 x_2 \bar{x}_1 \bar{x}_0$	$\bar{x}_3 x_2 x_1 \bar{x}_0$
$\bar{x}_3 \bar{x}_2$	X	X		X
$\bar{x}_3 x_2$			1	
$x_3 \bar{x}_2$	1	X	X	X
$x_3 x_2$	1	1		1

$$B_2 = \bar{x}_2 \bar{x}_1 + \bar{x}_0 \bar{x}_2 + x_2 x_1 x_0$$

B_3

$x_3 \backslash x_2 \backslash x_1 \backslash x_0$	$\bar{x}_3 \bar{x}_2 \bar{x}_1 \bar{x}_0$	$\bar{x}_3 \bar{x}_2 x_1 \bar{x}_0$	$\bar{x}_3 x_2 \bar{x}_1 \bar{x}_0$	$\bar{x}_3 x_2 x_1 \bar{x}_0$
$\bar{x}_3 \bar{x}_2$				
$\bar{x}_3 x_2$				
$x_3 \bar{x}_2$				
$x_3 x_2$				

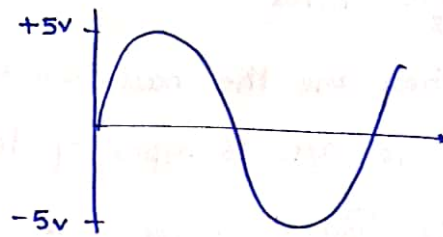
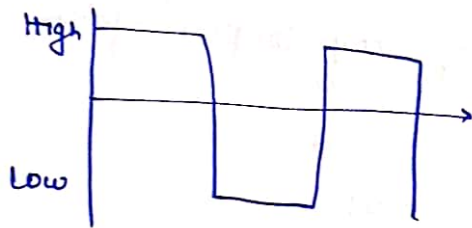
$$B_3 = \bar{x}_1 x_0 x_3$$

Electronics System & Circuits

Digital Electronics

Analog Electronics

- In Analog Electronics, Voltage may take any of the infinite possible value whereas in Digital Electronics we consider only two levels - logic high and logic low.



- Both Analog and Digital Electronics use same components.
- Digital circuit is also called logic circuit or switching ckt.
- Transition time of digital system is assumed to be zero.

Advantages of Digital Electronics

- Designing of digital ckt is easy.
- Information storage is easy.
- Effect of noise is less.
- Digital circuits are more versatile (reusable)

Application of DE :-

- It is preferred in field of measurement, control system, comm., and data processing over analog electronics.

Disadvantage of Digital Electronics :-

- Real world is analog. Hence required A/D, D/A in the system.

* Designing of Digital System

Step 1:- System Designing : Complete System divided into various block and their interconnection

Step 2:- Logic Designing : Each block Design using logic gate

Step 3:- Circuit Designing : Technology used to design gate

* Logic Gates

→ They are the basic building block of any Digital System

There are 3 types of Logic Gates

i) Basic Gates → AND OR NOT

ii) Universal Gates → NAND NOR

iii) Special purpose Gates → Ex-OR Ex-NOR

I. Basic Gates

i) AND Gate

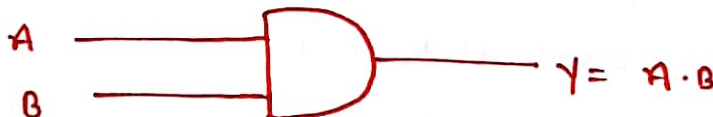


Fig. Logic Symbol of 2 i/p AND gate

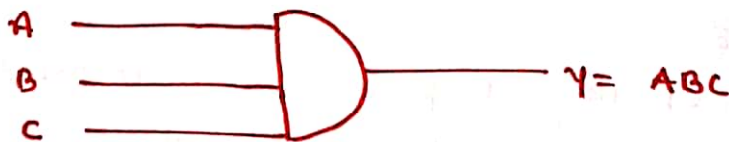


Fig. Logic Symbol of 3 i/p AND gate

AND gate is a logic gate whose o/p is high only when all the input are high

OR AND gate is a logic gate whose o/p is logic low if atleast one of the input is at logic low

Truth table :-

A	B	C	$Y = ABC$
L	L	L	L
L	L	H	L
L	H	L	L
L	H	H	L
H	L	L	L
H	L	H	L
H	H	L	L
H	H	H	H

note:- Truth table is a table which consist of all the possible combination of i/p along with their corresponding o/p.

ii) OR Gate

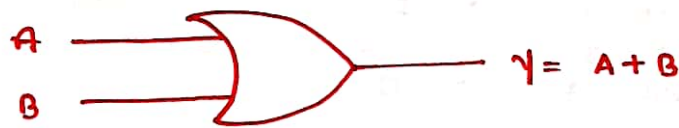


fig: Logic symbol of 2 i/p OR gate

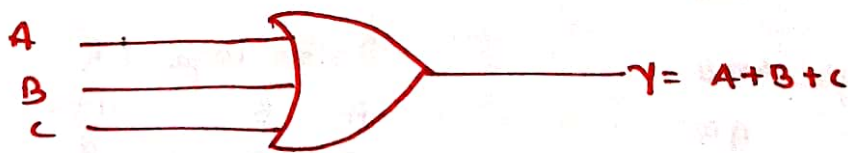


fig. 3 i/p OR gate

OR gate is a logic gate whose o/p is logic High when any one input is logic High

OR gate is a logic gate whose o/p is logic low only when all input are logic low.

Truth Table:-

A	B	C	$Y = A+B+C$
L	L	L	L
L	L	H	H
L	H	L	H
L	H	H	H
H	L	L	H
H	L	H	H
H	H	L	H
H	H	H	H

note:- Positive Logic Digital System

High $\longleftrightarrow$ '1'

Low $\longleftrightarrow$ '0'

Negative Logic Digital System

High $\longleftrightarrow$ '0'

Low $\longleftrightarrow$ '1'

Positive Logic AND

A	B	AB
0	0	0
0	1	0
1	0	0
1	1	1

Negative Logic AND

A	B	AB
1	1	1
1	0	0
0	1	0
0	0	0

Positive Logic OR

A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

Negative Logic OR

A	B	A+B
1	1	1
1	0	0
0	1	0
0	0	0

Key point :- Positive logic $A \cdot B \equiv$ Negative logic OR
Negative logic $A \cdot B \equiv$ Positive logic OR

III) NOT gate



Fig. Logic Symbol of NOT gate

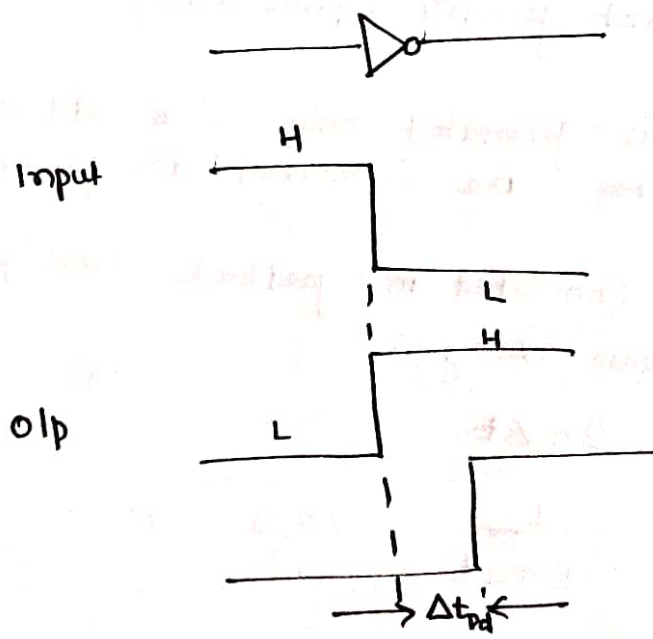
NOT gate has only one input

Truth Table

A	$Y = \bar{A}$
0	1
1	0

NOT gate is a logic gate whose output is Complement of its input.

Propagation Delay :- Time taken by the gate to reflect the change in input to the output



where, Δt_{pd} is delay time.

ex. Consider a Circuit

i)

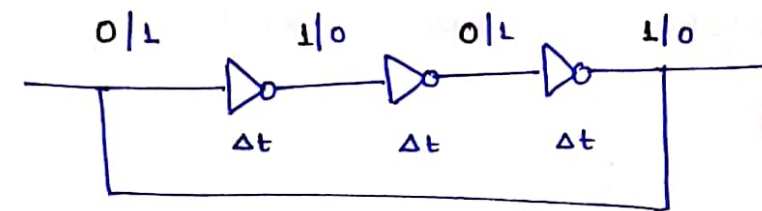
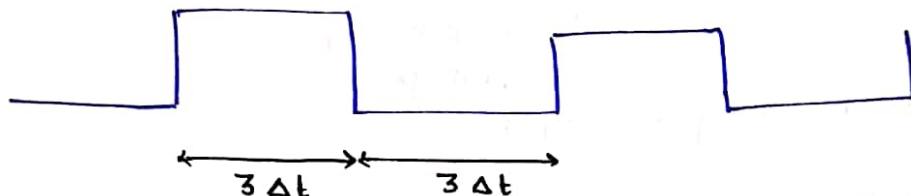


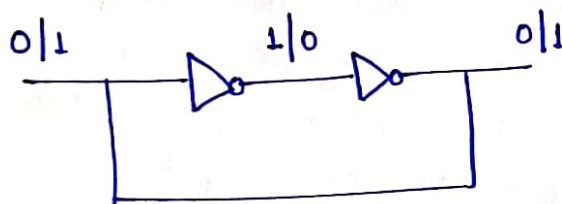
fig. Square wave
Generator
OR
Clock generator
OR
Ring Oscillator
OR
Astable Multi
vibrator



$$T = 6\Delta t$$

$$f = \frac{1}{6\Delta t}$$

ii)



It will not generate Square waveform

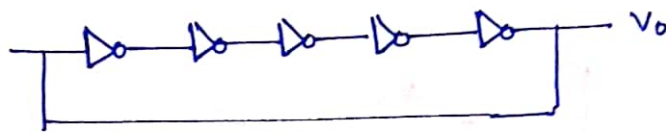
note:- Square Wave is generated only when odd no. of Cascaded Inverters are Connected in feed back.

For 'n' Inverters Connected in feed back time period of generated Square waveform

$$T = 2n\Delta t$$

$$f = \frac{1}{2n\Delta t}$$

Que. For the Ring Oscillator shown in fig the propagation Delay of each Inverter is 100ps . What is fundamental frequency of the oscillator o/p

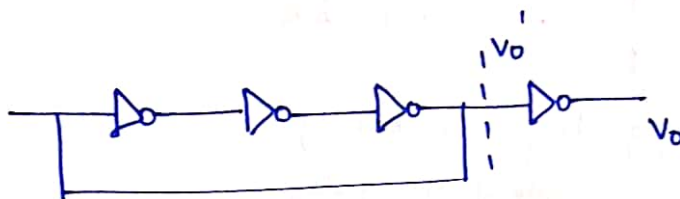


Sol

$$n = 5$$

$$f = \frac{1}{2 \times 5 \times 100} = 1\text{GHz}$$

Que.



Propagation delay of each not gate is 2ns . The Time period of generated Square wave is

Sol

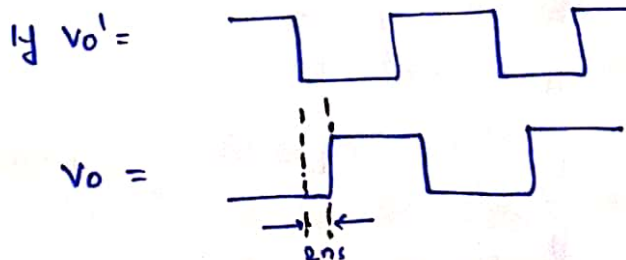
$$T = 6\Delta t + \Delta t$$

$$= 12\text{ns} + 2\text{ns}$$

$$= 14\text{ns}$$

Wrong

Shifting does not affect time period



$$T = 6\Delta t$$

$$= 12\text{ns} //$$

Universal Gate:-

1) NAND gate:- It is an AND gate followed by a nor gate



OR

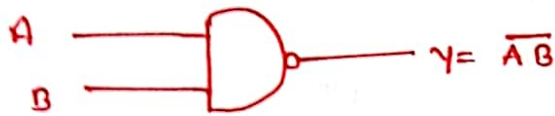


fig. Logic Symbol of 2/p NAND gate

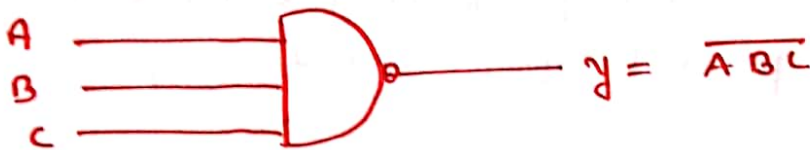


fig. Logic Symbol of 3/p NAND gate

Truth Table:

A	B	C	A B C	$Y = \overline{ABC}$
L	L	L	L	H
L	L	H	L	H
L	H	L	L	H
L	H	H	L	H
H	L	L	L	H
H	L	H	L	H
H	H	L	L	H
H	H	H	H	L

NAND gate is a logic gate whose output is logic low only when all the inputs are at logic high.

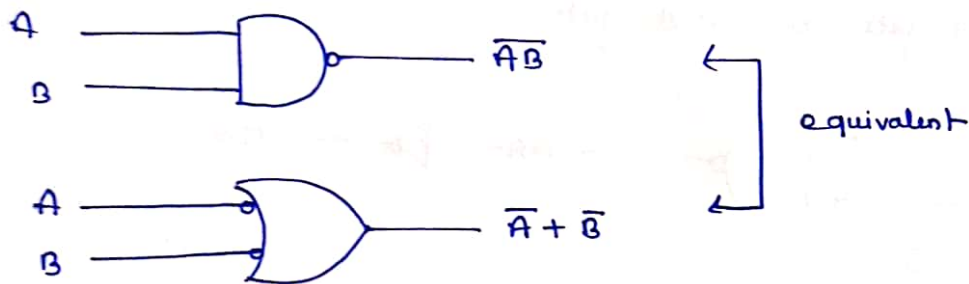
OR

NAND gate is a logic gate whose output is logic high when any of the inputs is at logic low.

ex.

A	B	$\overline{AB}$	$\overline{A}$	$\overline{B}$	$\overline{A} + \overline{B}$
0	0	1	1	1	1
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	0	0	0

$$\overline{AB} = \overline{A} + \overline{B}$$

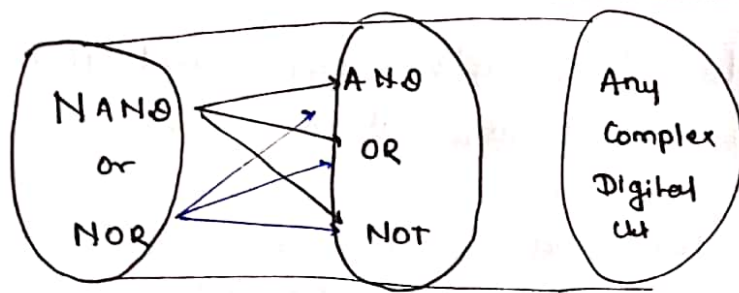


Key point: NAND gate is bubbled OR gate.

bubbled OR gate is active low OR gate or negative OR gate
Normal OR gate is active High OR gate or

Basic Gates :- AND, OR, NOT are called as basic gates as we can implement a digital circuit of any complexity using combination of these basic gates.

Universal Gate :- NAND & NOR gates are called as Universal gates as these gates (either NAND or NOR gates) are single handedly capable of designing a digital circuit of any complexity.



Que. Justify NAND gate as NOT gate

Sol

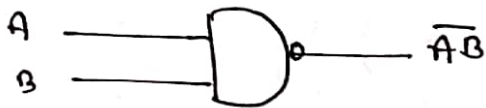


fig. NAND gate as NOT gate

NAND gate as AND gate



fig. NAND gate as AND gate

NAND gate as OR gate

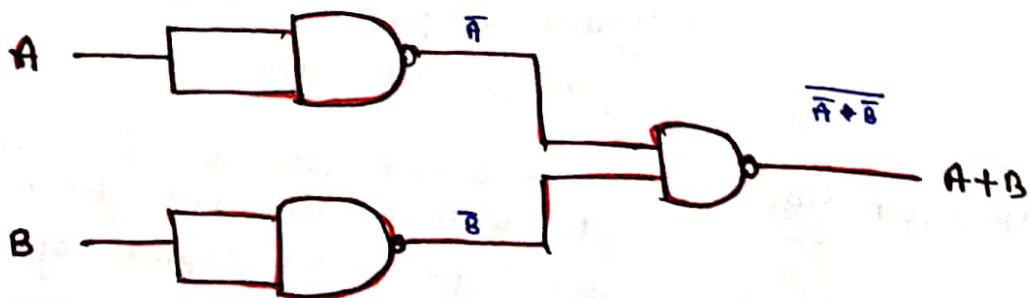


fig. NAND gate as OR gate

$$\text{Required function} = A+B = \overline{\overline{A+B}} \\ = \overline{\overline{A} \cdot \overline{B}}$$

Que. find the Min. no. of two input NAND gate required to design the following functions

① $F_1 = A+B$

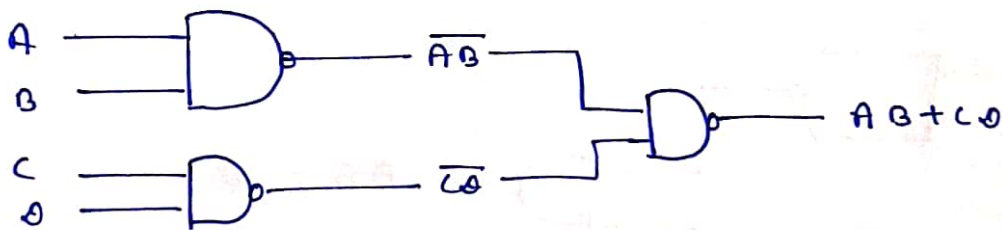
② $F_2 = AB+CD$

③ $F_3 = A+BC$

Sol 1:- Step 1: $A+B = \overline{\overline{A+B}}$

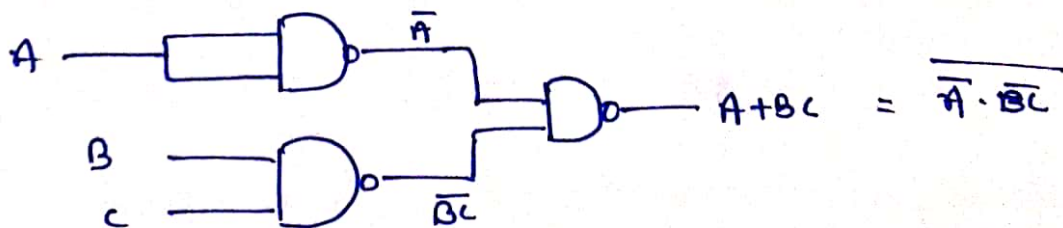
Step 2: De Morgan Law $\overline{\overline{A+B}} = \overline{\overline{A} \cdot \overline{B}}$
 $= A+B$

Sol 2: $AB+CD = \overline{\overline{AB+CD}}$
 $= \overline{\overline{AB} \cdot \overline{CD}}$

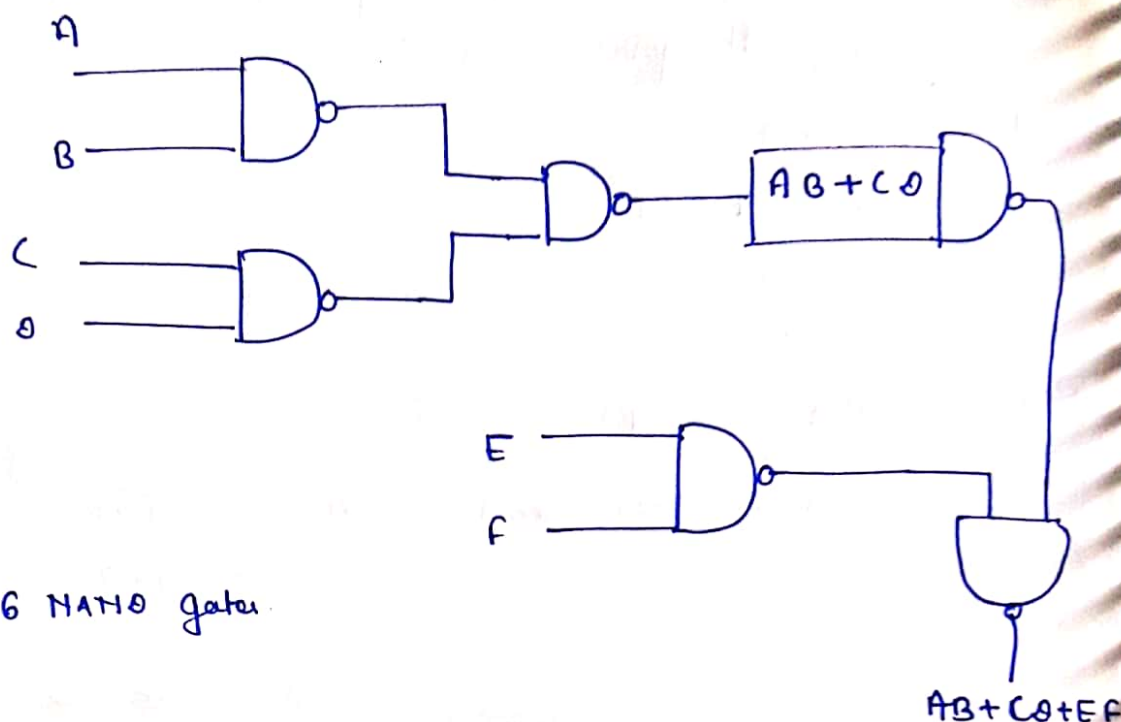


Sol 3: $F_3 = A+BC$

$$= \overline{\overline{A+BC}} \\ = \overline{\overline{A} \cdot \overline{BC}}$$

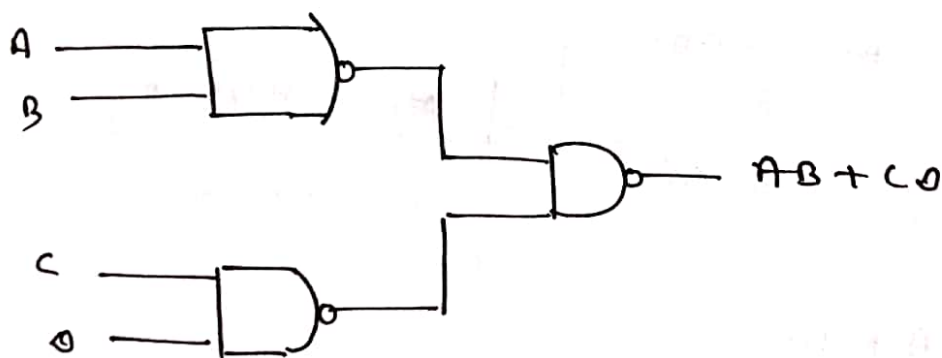


④ $F = AB + CD + EF$



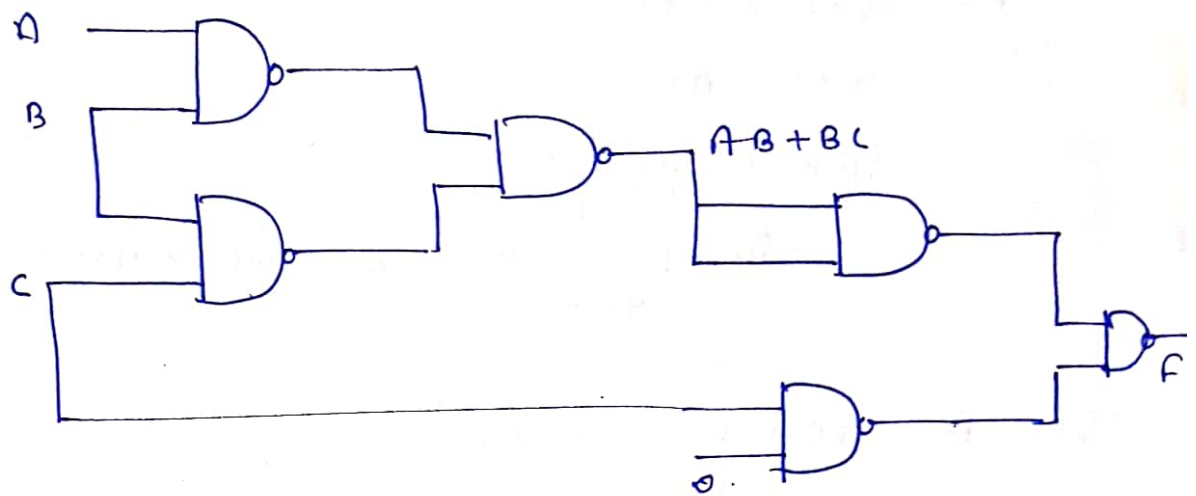
note:

$F = AB + CD$ form



Try to adjust in form of $AB + CD$

Q. $F = AB + BC + CD$

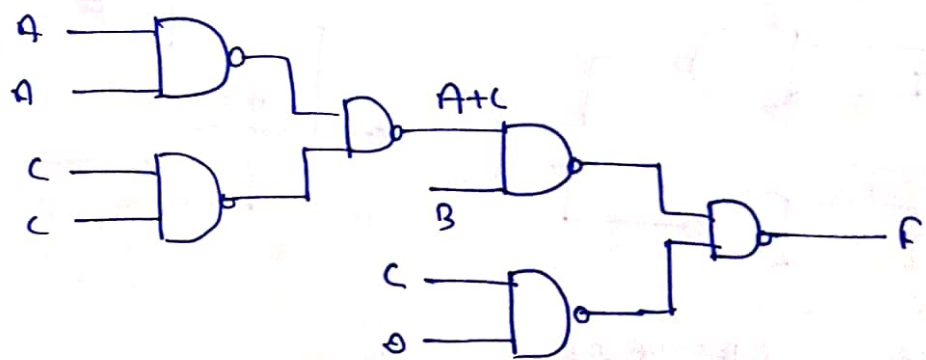


OR

$F = AB + BC + CD$

$= B(A + C) + CD$

$= B(AA + CC) + CD$



Que. $F = AB + BC + CD + DA$

$= (A + C)B + D(C + A)$

$= \frac{(AA + CC)B}{X} + \frac{D(CC + AA)}{Y}$

$X = 3 \text{ NAND gates}$

$Y = 3 \text{ NAND gates}$

now $F = BX + DY$

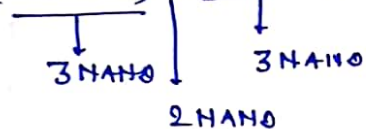
$F = 3 \text{ NAND gates}$ So total 9

OR $F = AB + BC + CA + DA$

$$F = B(A+C) + D(A+C)$$

$$= (B + 0)(A + C)$$

$$= (B \cdot B + D \cdot D) \cdot (\underbrace{A \cdot A + C \cdot C})$$



= Total 8 NAND gate

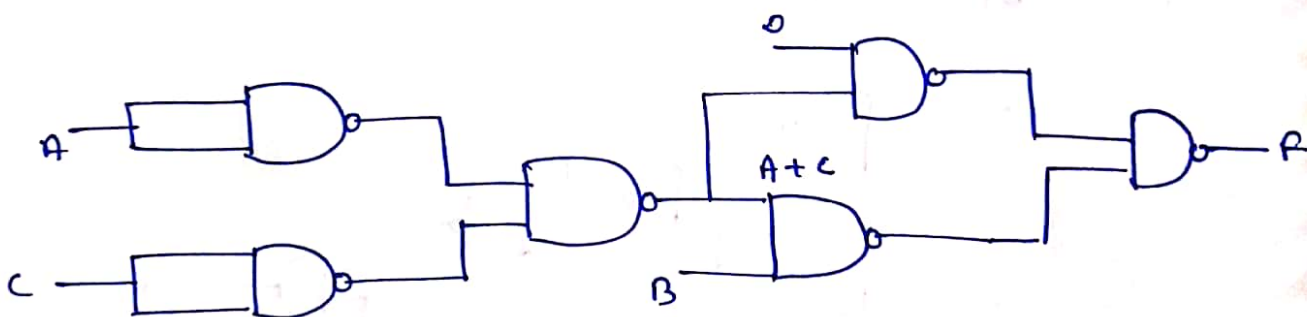
OR $F = AB + BC + CD + DA$

$$= B(A+C) + D(A+C)$$

$$= \underbrace{3x + 9x}_{3 \text{ HAD}}$$

and $A + C = AA + CC = 3 \text{ NaHCO}_3$

So total 6 NAND gate //



7. $f = ABC + DE$

$$= \overline{A \cdot B \cdot C} + \overline{A \cdot B \cdot C} \cdot \overline{D \cdot E}$$

$$= A B C + D E$$

$$= \underbrace{A \times + 05}$$

3 gate + 2 gate (x)

= 5 gate

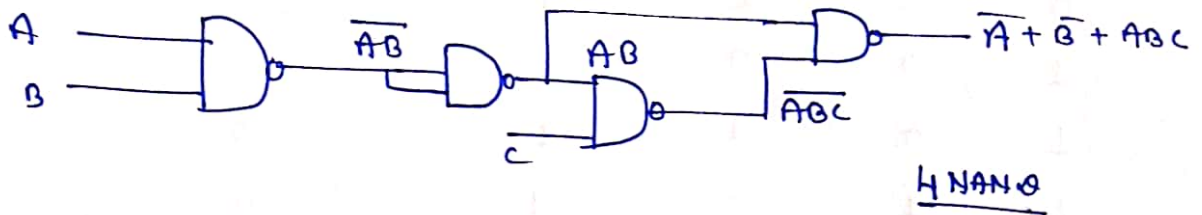
8. $\overline{A} + \overline{B} + ABC$

Sol $\overline{A} \cdot \overline{A} + \overline{B} \cdot \overline{B} + A \cdot X$

$$F = \overline{A} + \overline{B} + ABC$$

$$= \overline{AB} + \underline{ABC}$$

1 gate



$$F = \overline{A} + \overline{B} + ABC$$

$$= \overline{AB} + ABC$$

$$= \overline{\overline{AB} + ABC}$$

$$= \overline{AB \cdot \overline{ABC}}$$

II. NOR gate:- NOR gate is an OR gate followed by a NOT gate

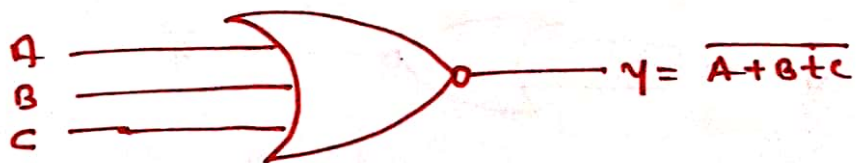
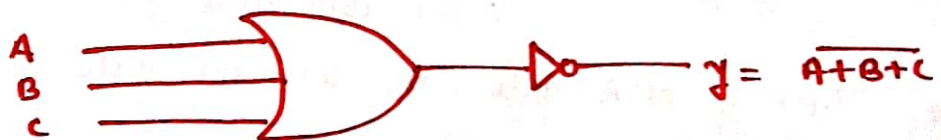


fig. Logic diag. of 3 input NOR gate

Truth table:

A	B	C	$A+B+C$	$\overline{A+B+C}$
0	0	0	0	1
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	1	0
1	0	1	1	0
1	1	0	1	0
1	1	1	1	0

NOR gate is a logic gate whose o/p is logic high only when all the i/p are logic low

OR

NOR gate is a logic gate whose o/p is logic low if any of its input is logic high.

Note: NOR gate $\equiv$ Bubbled AND gate / Negative AND Gate / Active low AND gate

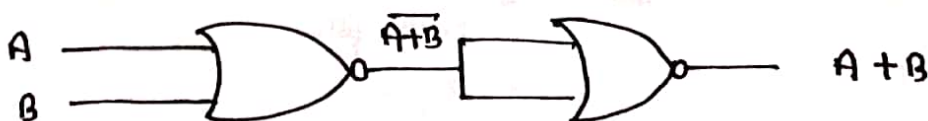
Que. Justify NOR gate as universal gate

i) NOT gate



$$\left\{ \begin{array}{l} \overline{A+A} = \bar{A} \\ A+\bar{A} = A \end{array} \right.$$

ii) OR gate



iii) AND gate

$$\begin{aligned} Y &= A \cdot B \\ &= \overline{\overline{A \cdot B}} \\ &= \overline{\overline{A} + \overline{B}} \end{aligned}$$

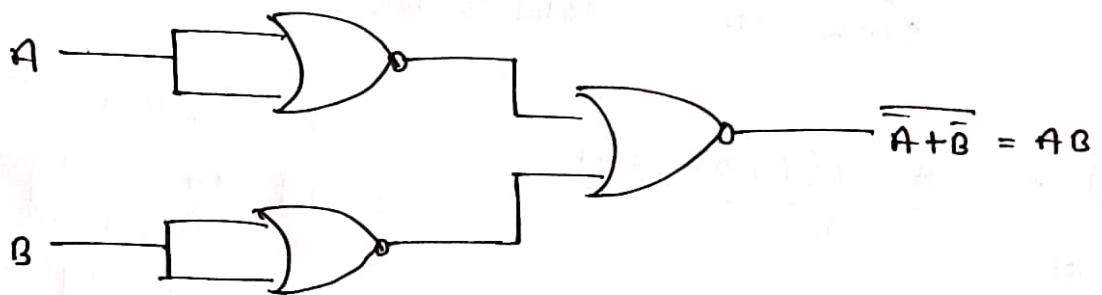
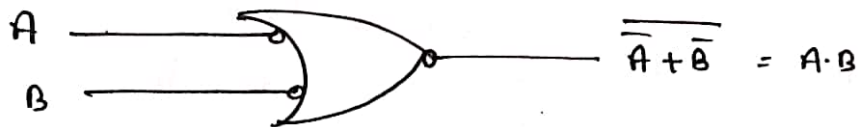
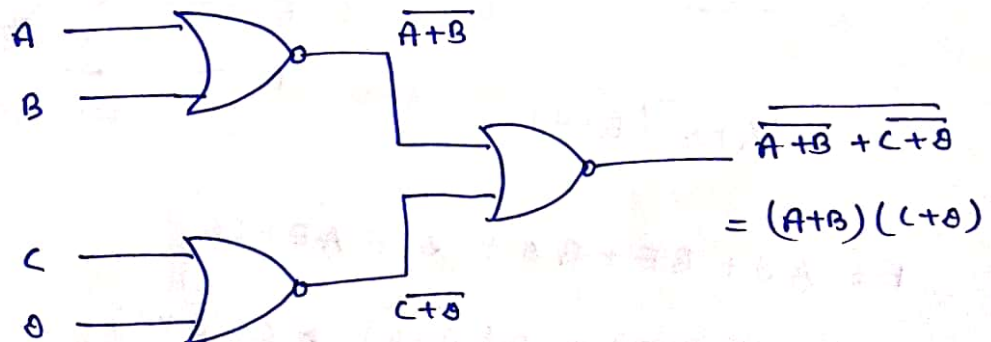


fig. AND gate using NOR

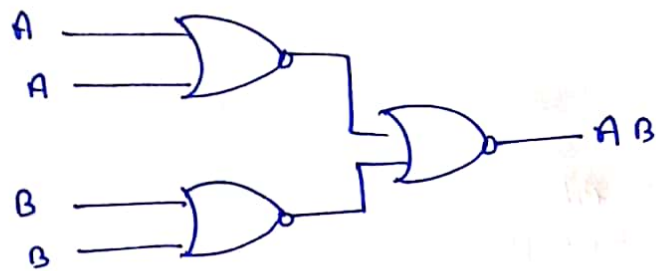
Note :



Que1. Design following function using Min no. of 2 input NOR gates.

i) $F = A \cdot B$

Sol $F = A \cdot B = (A+A) \cdot (B+B)$



ii) $F = A(B + C)$

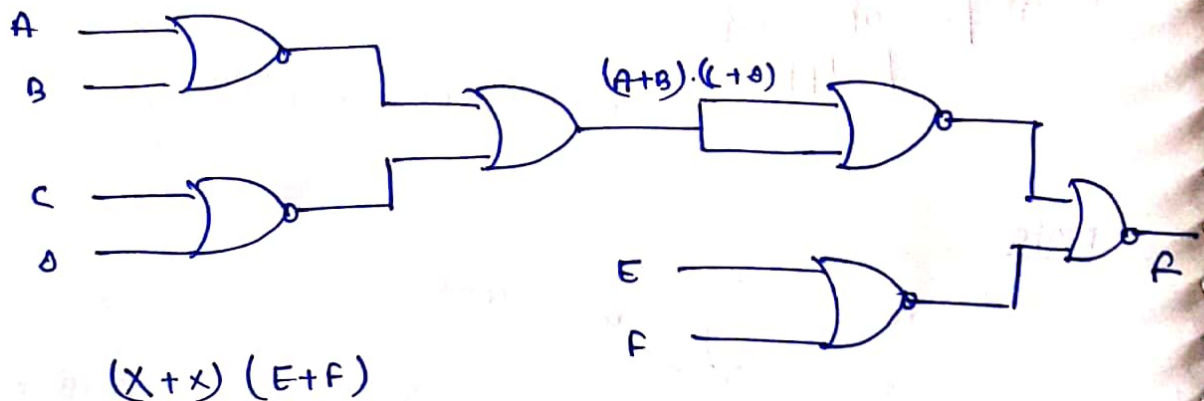
$$= (A + A) \cdot (B + C)$$

↓ ↓
1 NOR 1 NOR

Total 3 NOR

iii) $F = \frac{(A+B)(C+D)(E+F)}{X}$

Sol



iv) $F = A\delta + BE + B\delta + C\delta + AE + CE$

$$= A(A+B) + E(A+B) + C(\delta + E)$$

$$= (A+B)(\delta + E) + C(\delta + E)$$

$$= (\delta + E)(A+B) + (C+C)(\delta + E)$$

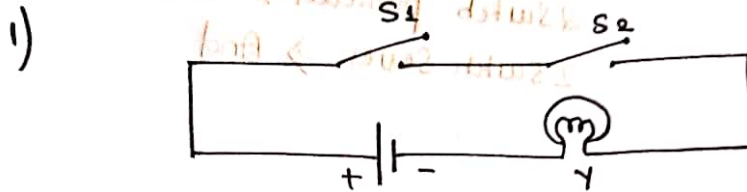
$$= (A+B)X + X(C+C)$$

$$= (A+B)(X+X) + (X+X)(C+C)$$

$$F = (A+B+C)(D+E)$$

$$= \underbrace{(X+C)}_{2 \text{ gate}} \underbrace{(D+E)}_{3 \text{ gate}} = 5 \text{ gate}$$

* Switching Circuit representation of logic gates :-



Switch ON $\Rightarrow$ Short circuited i.e high

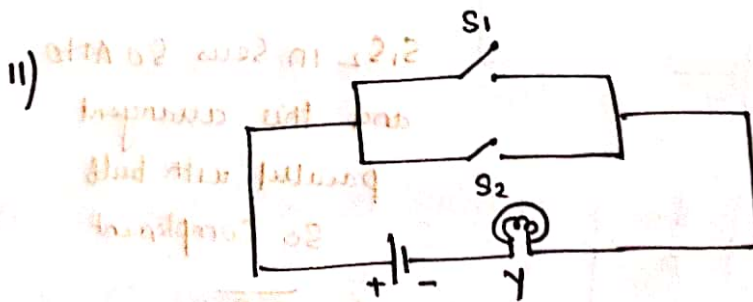
Switch OFF $\Rightarrow$ open circuited i.e low

Bulb glowing $\Rightarrow$ High

Bulb not glowing $\Rightarrow$ low

S ₁	S ₂	Y
OFF 0	OFF 0	no glow 0
OFF 0	ON 1	not glow 0
ON 1	OFF 0	not glow 0
ON 1	ON 1	glow 1

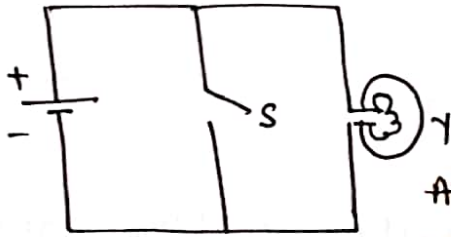
Hence this Switching ckt represents AND gate.



S ₁	S ₂	Y
0	0	0
0	1	1
1	0	1
1	1	1

It Represent OR gate

iii)



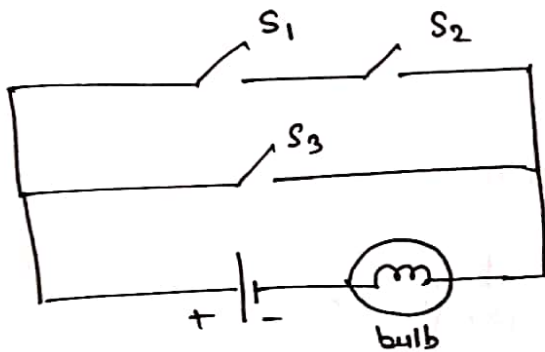
~~AND~~ → Series with bulb
~~NAND~~ → parallel with bulb

2 switch parallel ⇒ OR
 2 switch series ⇒ And

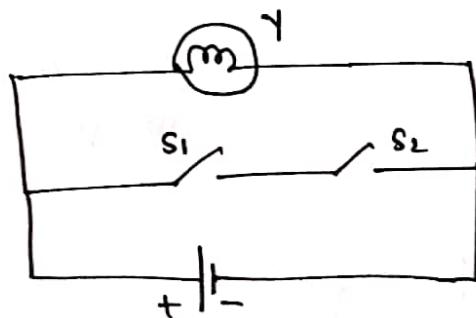
S	Y
0	1
1	0

This is NOT gate

iv)



→ $S_3 + S_1 S_2$
 and this arrangement
 series with bulb
 Hence no complement
 applied



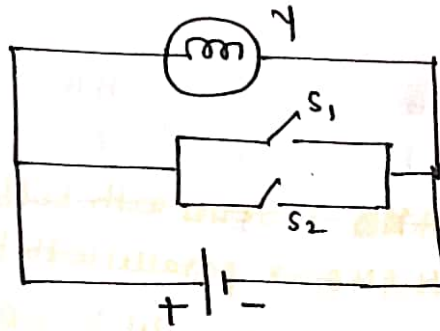
This is NAND gate

$S_1 S_2$ in series so AND
 and this arrangement
 parallel with bulb
 so complement

S ₁	S ₂	Y
0	0	1
0	1	1
1	0	1
1	1	0

$$Y = \overline{S_1 S_2}$$

v)

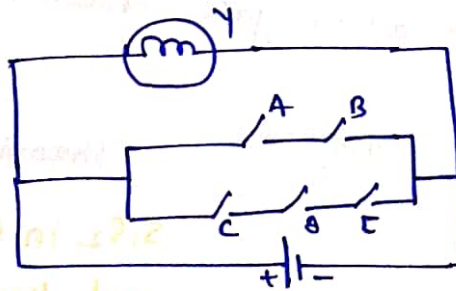


S_1	S_2	Y
0	0	1
0	1	0
1	0	0
1	1	0

This arrangement is NOR gate

Que. Draw the Truth table & Switching circuit.

$$F = \overline{AB + CDE}$$



Parallel with bulb
be Complement
applied.

* Special Purpose gates :-

1) Exclusive OR gate (Ex-OR) :-

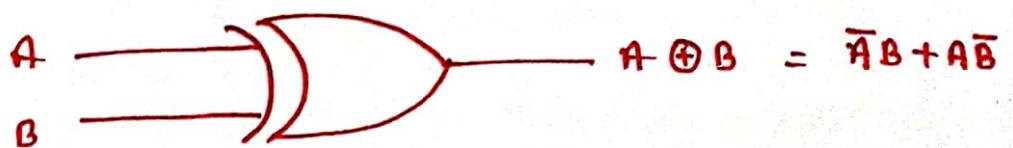


fig. Logic Symbol of
2 i/p ex-or gate

Truth table

A	B	$\bar{A}$	$\bar{B}$	$\bar{A}B$	$A\bar{B}$	$\bar{A}B + A\bar{B}$
0	0	1	1	0	0	0
0	1	1	0	1	0	1
1	0	0	1	0	1	1
1	1	0	0	0	0	0

both i/p
different

0	1
1	0

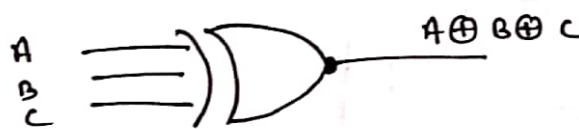
Def.:- Ex-OR gate is a logic gate whose o/p is logic high when both the input are unequal or different

OR

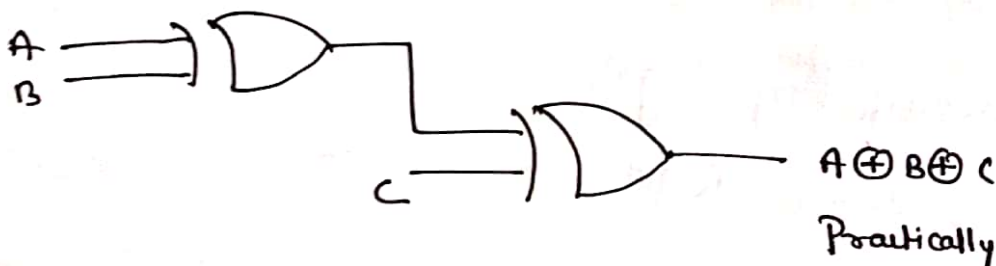
Ex-OR gate is a logic gate whose o/p is logic low when both the input are similar or equal.

- Ex-OR gate detects unequal inputs by producing logic high at the output so it is also called inequality detector.

note:- 3 or more input ex-or gate does not exist.



Theoretically Correct



Practically

ex.

A	B	C	$A \oplus B$	$(A \oplus B) \oplus C$	No. of 1's in A, B & C
0	0	0	0	0	even
0	0	1	0	1	→ odd
0	1	0	1	1	→ odd
0	1	1	1	0	even
1	0	0	1	1	→ odd
1	0	1	1	0	even
1	1	0	0	0	even
1	1	1	0	1	→ odd

Ex-OR gate is an odd number of 1's detector

ex. $1 \oplus 0 \oplus 1 \oplus 0 \oplus 0 \oplus 1 = 1$

Que. find

- 1) $A \oplus 0 = A$
- 2) $A \oplus 1 = \bar{A}$
- 3) $A \oplus A = 0$
- 4) $A \oplus \bar{A} = 1$

Que. 14 $A \oplus B = C$

$A \oplus C = ?$

Sol $\bar{A}B + A\bar{B} = C$

$A \oplus C$

$\Rightarrow A \oplus A \oplus B$

$\Rightarrow 0 \oplus B$

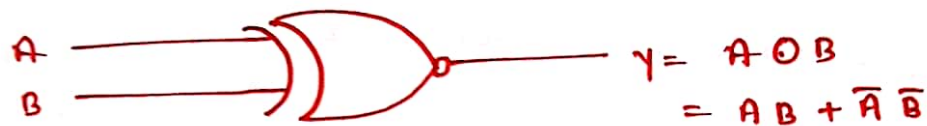
$= B \quad //$

5) $A \oplus B = C$ Note
 $A \oplus C = B$

6) If $A \oplus B = C$
 then $A \oplus B \oplus C =$

Sol $C \oplus C = 0$
 $A \oplus B \oplus C = 0$

11) Ex-NOR gate :-



Ex-nor gate is an ex-or gate followed by an inverter.

	A	B	$\bar{A}$	$\bar{B}$	$A \cdot B$	$\bar{A} \cdot \bar{B}$	$AB + \bar{A}\bar{B}$
both i/p equal	0	0	1	1	0	1	1
	0	1	1	0	0	0	0
	1	0	0	1	0	0	0
both i/p equal	1	1	0	0	1	0	1

dy. Output is Logic high when both Input are Similar
 in Case of ex-nor gate.

o/p is Logic low when both i/p are different
 in Case of ex-nor gate.

Ex-nor gate is also Called equality detector
 because it give o/p logic high when both inputs
 are equal.

$$\begin{aligned}
 A \oplus B \oplus C &= A \oplus B \cdot \bar{C} + C \cdot \overline{A \oplus B} \\
 &= \overline{A \oplus B} \cdot \bar{C} + C \cdot \overline{A \oplus B} \\
 &= \overline{A \oplus B} \cdot \bar{C} + C \cdot A \oplus B \\
 &= \bar{x} \bar{y} + xy \\
 &= x \odot y \\
 &\Rightarrow A \odot B \odot C
 \end{aligned}$$

Hence

$$A \oplus B \oplus C = A \odot B \odot C$$

Concept:-

$$\overline{A \oplus B} = A \odot B$$

$$A \oplus B \oplus C = A \odot B \odot C$$

$$\overline{A \oplus B \oplus C \oplus D} = A \odot B \odot C$$

$$\begin{aligned}
 \text{XOR} &= \overline{\text{XNOR}} ; \text{ for even} \\
 &\quad \text{no. of Input} \\
 \text{XOR} &= \text{XNOR} ; \text{ for odd no.} \\
 &\quad \text{of I/p}
 \end{aligned}$$

for odd no. of Input

$$\text{XOR} = \text{XNOR} \quad \text{i.e. both XNOR is odd no. of 1's detector if I/p are odd}$$

for even no. of Input

$$\overline{\text{XOR}} = \text{XNOR} \quad \text{i.e. XNOR is even detector if I/p are even}$$

eg 1. $1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 1 = \text{even 1's} \quad \text{o/p} = 0$

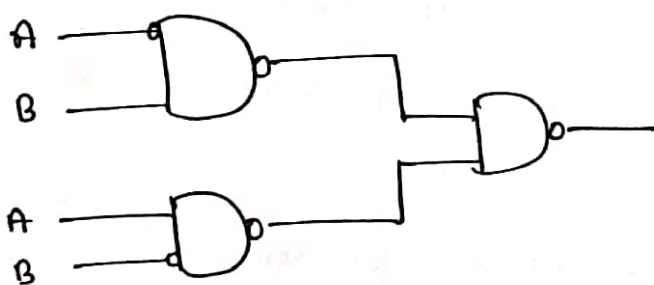
ii) $1 \oplus 1 \oplus 0 \oplus 1 \oplus 1 \oplus 1 \oplus 0 = \text{odd 1's} \quad \text{o/p} = 1$

iii) $1 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 = \text{even 1's} \text{ and } \text{even 1's} \quad \text{o/p} = '1'$

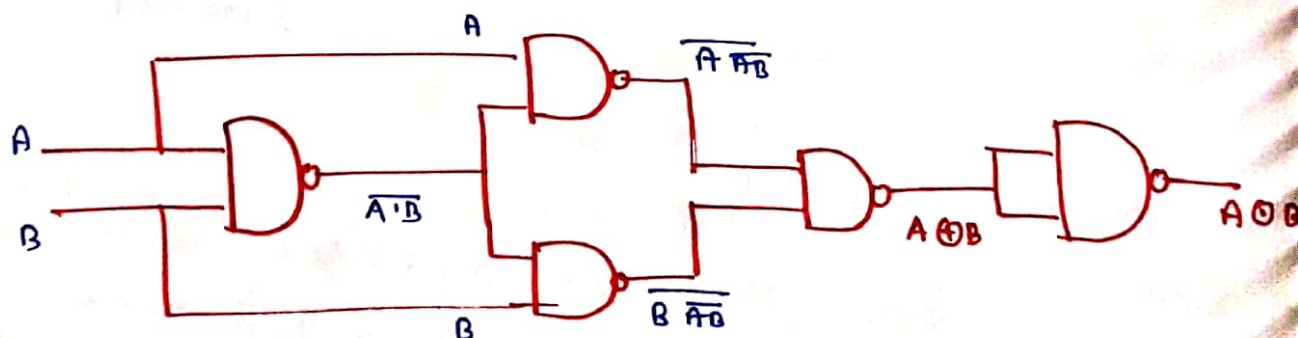
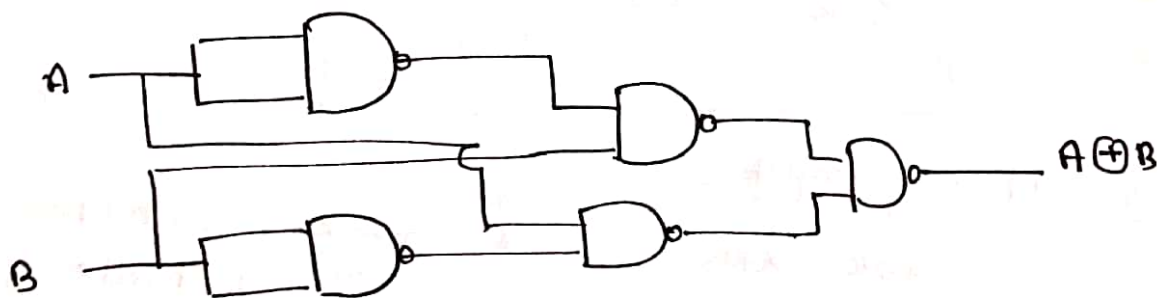
iv) $1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \oplus 0 \oplus 1 = \text{odd 1's and odd 1's} \quad \text{o/p} = '1'$

Que. find Min. two i/p NAND gate to design an XOR fun.

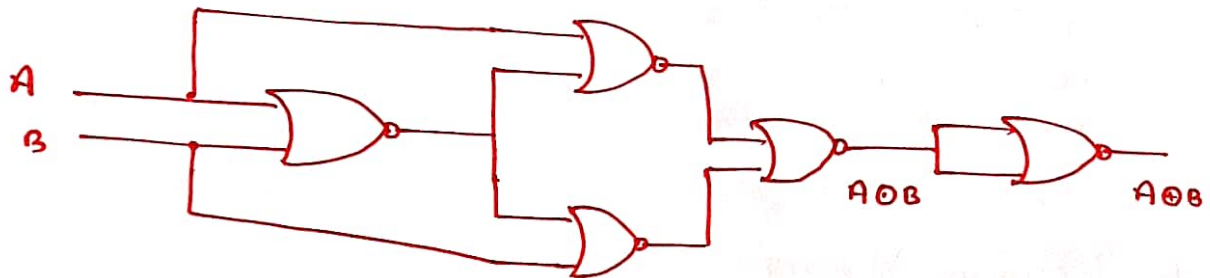
Sol $A \oplus B$
 $= \bar{A}B + A\bar{B}$



Hence total 5 NAND gate



- 4 NAND gate required to design XOR function
 - 5 NAND gate required to design XNOR function
- Q40. Design XOR gate using NOR gate



- 4 NOR gate Required to design XNOR function
- 5 NAND gate required to design XOR function

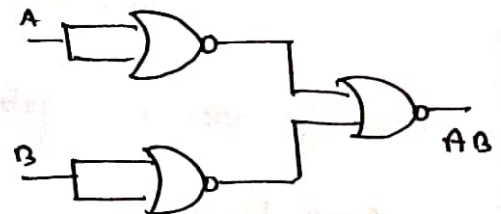
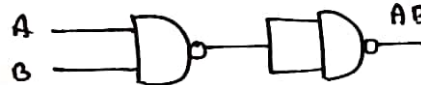
NAND Gate

NOR Gate

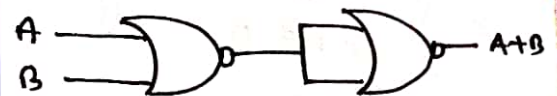
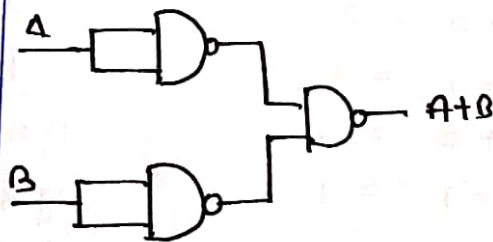
i) NOT



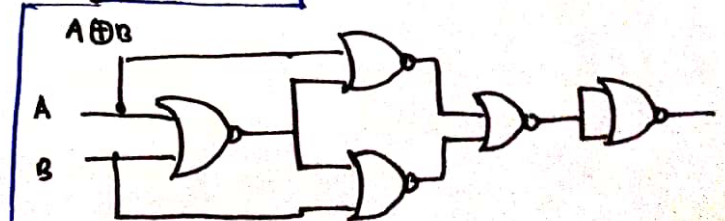
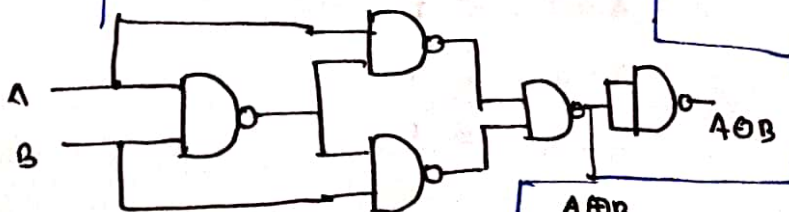
ii) AND



iii) OR



iv) XOR



No. of Gates

	NAND	NOR
NOT	1	1
AND	2	3
OR	3	2
XOR	4	5
XNOR	5	4

* Boolean Algebra

Binary

$$\begin{array}{r} 1 \\ + 1 \\ \hline 0 \ 1 \end{array}$$

Decimal

$$\begin{array}{r} 1 \\ + 1 \\ \hline 2 \end{array}$$

Both Numerical
Significance
but not logical

Boolean

$$\begin{array}{r} 1 \\ + 1 \\ \hline 1 \end{array}$$

Logical Significance

Laws of Boolean Algebra

AND laws

$$\begin{aligned} 0 \cdot 0 &= 0 \\ 0 \cdot 1 &= 0 \\ 1 \cdot 0 &= 0 \\ 1 \cdot 1 &= 1 \\ A \cdot 0 &= 0 \\ A \cdot 1 &= A \\ A \cdot A &= A \\ A \cdot \bar{A} &= 0 \end{aligned}$$

OR laws

$$\begin{aligned} 0 + 0 &= 0 \\ 0 + 1 &= 1 \\ 1 + 0 &= 1 \\ 1 + 1 &= 1 \\ A + 0 &= A \\ A + 1 &= 1 \\ A + A &= A \\ A + \bar{A} &= 1 \end{aligned}$$

NOT laws

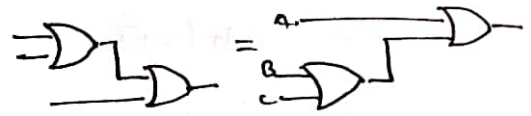
$$\begin{aligned} \bar{0} &= 1 \\ \bar{1} &= 0 \\ \text{If } A=0 \quad \bar{A} &= 1 \\ \text{If } A=1 \quad \bar{A} &= 0 \\ \overline{\bar{A}} &= A \end{aligned}$$

Commutative Law

- i) $A \cdot B = B \cdot A$
- ii) $A + B = B + A$
- iii) $\overline{A \cdot B} = \overline{B \cdot A}$
- iv) $\overline{A + B} = \overline{B + A}$
- v) $A \oplus B = B \oplus A$
- vi) $A \odot B = B \odot A$

Associative Law

i) $(A + B) + C = A + (B + C)$ OR



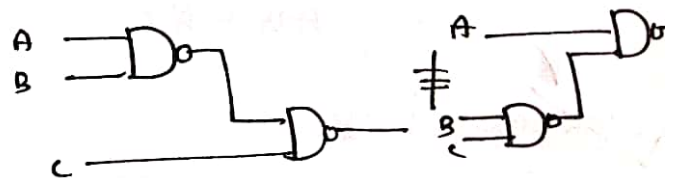
ii) $(A \cdot B) \cdot C = A \cdot (B \cdot C)$ AND

iii) $(A \oplus B) \oplus C = A \oplus (B \oplus C)$ XOR

iv) $(A \odot B) \odot C = A \odot (B \odot C)$ XNOR

v) $\overline{(A \cdot B) \cdot C} \neq \overline{A \cdot (B \cdot C)}$ NAND

NAND and NOR do not follow associative Law.



Distributive Law

1) $A(B + C) = AB + AC$

2) $A + BC = (A + B)(A + C)$

Absorption Law

1) $A + AB = A$

2) $A(A + B) = A$

Que. Simplify

$$\begin{aligned} F &= AB + \bar{B}C + AC \\ &= \cancel{AB(C+\bar{C})} + \bar{B}C(\cancel{A+\bar{A}}) + \cancel{AC(B+\bar{B})} \quad \text{Wrong} \\ &\Rightarrow \overline{AB + \bar{B}C + AC} \\ &> AB \end{aligned}$$

Sol :-

$$\begin{aligned} &AB + \bar{B}C + AC \\ &AB + \bar{B}C + AC(B + \bar{B}) \\ &\Rightarrow AB + \bar{B}C + ACB + AC\bar{B} \\ &\Rightarrow AB(1+C) + \bar{B}C(1+A) \\ &\Rightarrow AB + \bar{B}C \end{aligned}$$

* Consensus Law

$$i) AB + BC + A\bar{C} = BC + A\bar{C}$$

Hence only one variable C present in two variable in both normal and complemented form. Thus in output we get only those two term having C and to get o/p $AB(C+\bar{C})$ is applied

$$ii) \bar{A}B + BC + AC = \bar{A}B + AC$$

$$iii) \bar{A}\bar{B} + B\bar{C} + \bar{A}\bar{C} = \bar{A}\bar{B} + B\bar{C}$$

$$* iv) (A+B)(B+C)(\bar{A}+C) = (A+B)(\bar{A}+C)$$

$$v) (A+\bar{B})(B+C)(A+C) = (A+\bar{B})(B+C)$$

Prove: $(A+B)(B+C)(\bar{A}+C) = (A+B)(\bar{A}+C)$

Sol LHS

$$(A+B)(B+C+0)(\bar{A}+C)$$

$$\rightarrow (A+B)(\underbrace{B+C}_x + \underbrace{A\bar{A}}_{y\bar{z}})(\bar{A}+C)$$

$$\rightarrow (A+B)(A+B+C)(\bar{A}+B+C)(\bar{A}+C)$$

$$\rightarrow (\underbrace{A+B+0}_{x+y})(\underbrace{A+B+C}_{x+z})(\bar{A}+B+C)(\bar{A}+C+0)$$

$$\rightarrow (A+B+0.C)(\bar{A}+C+0.B)$$

$$= (A+B)(\bar{A}+C) \quad //$$

* Demorgan Theorem

$$i) \overline{A+B} = \bar{A} \cdot \bar{B}$$

$$ii) \overline{A \cdot B} = \bar{A} + \bar{B}$$

$$iii) \overline{A+B+C+D+E+\dots} = \bar{A} \cdot \bar{B} + \bar{C} \cdot \bar{D} \dots$$

$$iv) \overline{A \cdot B \cdot C \cdot D \cdot E \dots} = \bar{A} + \bar{B} + \bar{C} \dots$$

* Principle of Duality

$$i) '0' \longleftrightarrow '1'$$

$$ii) '1' \longleftrightarrow '0'$$

iii) Do not Complement Variable

eg. 1) $F = AB + CD$

$F = A \cdot B + C \cdot D$

$F^0 = (A+B) \cdot (C+D)$

ii) $f_2 = A \cdot 1$

$f_2^0 = A + D$

Principle:- It states that "If we have a valid boolean eqⁿ then its dual eqⁿ is also valid"

eg. $A + BC = (A+B)(A+C)$

Apply dual

$A \cdot (B+C) = A \cdot B + A \cdot C$

eg. $A \cdot \bar{A} = 0$

$\xrightarrow{\text{dual}} A + \bar{A} = 1$

eg. $A + AB = A$

dual $A \cdot (A+B) = A$

Que. $F = AB + BC + AC$ find dual

$F^0 = (A+B)(B+C)(A+C)$

$= (AB + AC + B + BC)(A+C)$

$= AAB + AAC + AC + ABC + BAC + ACC + BC + BCC$

$= AB + AC + AC + ABC + AC + BC + B + ABC$

$= AB + AC + BC + ABC$

$= AB + BC + AC$

$F = F^0 \Rightarrow$ This type of function are called
Self dual function

Que. How many unique functions are possible using only two variables?

Sol

A	$\overline{A}B$	$A+\overline{B}$
B	$A\overline{B}$	$\overline{A}$
$\overline{A}B$	$A+B$	$\overline{B}$
$A\overline{B}$	$\overline{A+B}$	$\overline{A}B+A\overline{B}$
0	$\overline{A}\overline{B}$	$\overline{A}\overline{B}+AB$
1	$\overline{A+B}$	$A+\overline{A}\overline{B}$

By O/p combinations

A	B	$\gamma=0 \rightarrow AB$	$\overline{A}B+A\overline{B}$	$A+B$	$\overline{A+B}$	$\overline{A}\overline{B}$	$\overline{B}$	$\overline{A}$	$\overline{A}\overline{B}$
0	0	0	0	0	0	0	0	1	1
0	1	0	0	0	1	1	1	0	0
1	0	0	0	1	0	0	1	0	0
1	1	0	1	0	1	0	0	0	1

2^{2^n} unique functions can be formed by using n inputs.

• Total no. of self dual functions $2^{2^{n-1}}$

Quo. $F = A \oplus B$ find Complement & dual

$$\overline{F} = \overline{A \oplus B}$$

$$\overline{F} = \overline{A\overline{B} + A\overline{B}}$$

$$F^D = \overline{A}B + A\overline{B}$$

$$= (\overline{A}+B)(A+\overline{B})$$

$$= \overline{A}A + \overline{A}\overline{B} + AB + B\overline{B}$$

$$= \overline{A}\overline{B} + AB$$

Note:- XOR is such function $F^D = \overline{F}$

Que. $x = \{ (a+b\bar{c})d+e \} \bar{f} + g$

find $\bar{x}$

Sol $\{ (a+b\bar{c})d+e \} \bar{f} + g$

$\rightarrow \overline{\{ (a+b\bar{c})d+e \} \bar{f} + g}$

$\rightarrow \overline{(a+b\bar{c})d + e} + f * \bar{g}$

$\rightarrow \overline{(a+b\bar{c})d} * \bar{e} + f * \bar{g}$

$\rightarrow \{ \overline{(a+b\bar{c})} + \bar{d} \} \bar{e} + f * \bar{g}$

$\rightarrow \{ \bar{a} * \overline{b\bar{c}} + \bar{d} \} \bar{e} + f * \bar{g}$

$\rightarrow [\bar{a} \bar{e} \cdot \bar{e} (\bar{b} + c) + \bar{e} \bar{d} + f * \bar{g}] \bar{g} //$

$\rightarrow (\bar{a} \bar{b} \bar{e} + \bar{a} \bar{e} c + \bar{e} \bar{d} + f) \bar{g} \quad \text{--- (1)}$

find x^0

Sol $\{ a \cdot (b+\bar{c}) + d \cdot e \} + \bar{f} \cdot g$

$= [\{ a \cdot (b+\bar{c}) + d \} e + \bar{f}] g$

Individual Complement all variable

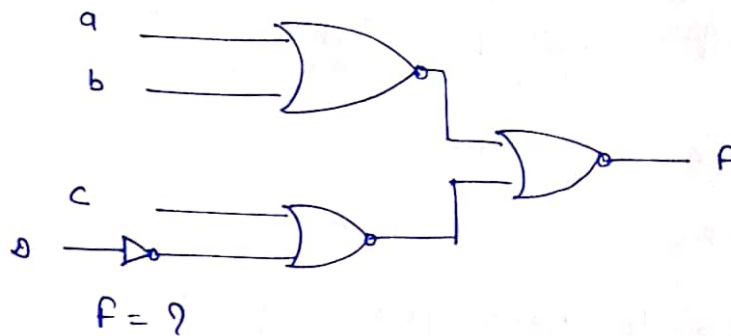
$[\{ \bar{a} \cdot (\bar{b} + c) + \bar{d} \} \bar{e} + f] \bar{g} \quad \text{--- (2)}$

eq (1) & (2) are Same Hence to find Complement

i) find dual

ii) Complement Individual

Que.



- a) $(a+b)(c+d)$
- b) $(\bar{a}+\bar{b})(c+d)$
- c) $(a+\bar{b})(c+\bar{d})$
- d) $(a+b)(\bar{c}+\bar{d})$

Que. What is the Minimum no. of gate required to Implement Boolean exp.
 $AB + c$ if we have to use only two i/p NOR.

Sol $F = AB + c$
 $= (A+B)(B+c)$
 $= 3 \text{ NOR.}$

Que. The Boolean function $y = AB + c$ is to be implemented using only two i/p NAND. The Min. no of gate required is 3.

Que. Assume that only X & Y logic input are available and their Complement $\bar{X}$ & $\bar{Y}$ are not available what is Min. no of two i/p NAND gate req. to Implement

$$X \oplus Y$$

Sol 4 NAND

Q4. The Boolean Exp. $Y = A + BC$ is to be realized using two 1/p gates of only one type. What is min. no. of gate required for the Implementation

Sol $Y = (A+B)(A+C)$ 3 NOR

OR $Y = A \cdot A + BC$ 3 NAND

Q5. Min. no of two 1/p NAND gate required to Implement the function

$$Z = A \bar{B} c$$

$$= \overline{\overline{A \bar{B} c}}$$

$$= \overline{A \bar{B} + \bar{c}}$$

$$= \bar{A} + B + \bar{c}$$

$$\begin{array}{r} \overline{A \bar{B} + \bar{c} \cdot \bar{c}} \\ \begin{array}{cc} \overline{A \bar{B}} & + & \overline{\bar{c} \cdot \bar{c}} \\ \text{X} & & \text{Y} \end{array} \\ \downarrow \\ \text{3 NAND} \\ \begin{array}{r} + 1 \bar{B} \\ + 1 A \bar{B} \\ + 1 \bar{c} \\ \hline \bar{c} \\ + 1 \\ \hline 7 \end{array} \end{array}$$

$$Z = A \bar{B} c$$

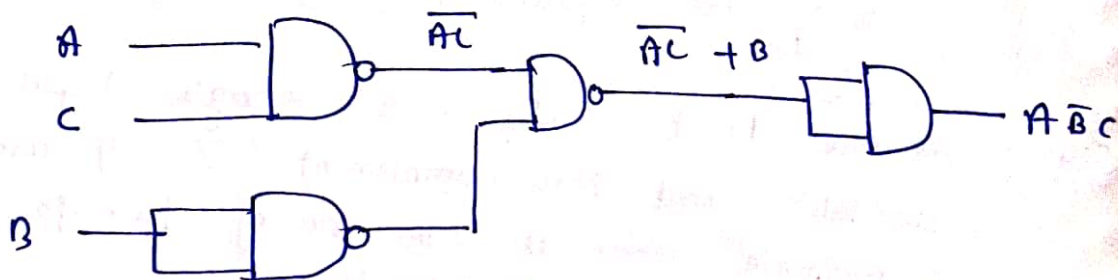
$$= \overline{\overline{A \bar{B} c}}$$

$$= \overline{A \bar{B} + \bar{c}}$$

=

$$\Rightarrow \overline{A \bar{c}} + \bar{B}$$

$$\overline{A \bar{c}} + B \cdot B$$



Que. $(X+Y)(X+\bar{Y}) + [(\overline{X\bar{Y}}) + \bar{X}]$

Sol $X + X\bar{Y} + XY + 0 + \overline{X\bar{Y}} \cdot X$

$\rightarrow X + XY + 0 + (\bar{X} + Y)X$

$\rightarrow X + 0 + XY$

$= X(1+Y)$

$= X //$

$X + Y\bar{X} + (\bar{X} + Y)X$

$X + XY$

$X //$

Que. $F(P, Q, R, S) = PQ + \bar{P}QR + \bar{P}Q\bar{R}S$

Minimal function F is

Sol $PQ + \bar{P}QR + \bar{P}Q\bar{R}S$

$\rightarrow PQ + \bar{P}Q(R + \bar{R}S)$

$\rightarrow Q[P + \bar{P}(R + \bar{R}S)]$

$\rightarrow Q[P + \bar{P}(R + \bar{R})(R + S)]$

$\rightarrow Q[P + \bar{P}(R + S)] \rightarrow Q[\bar{P}]$

$\rightarrow QP + Q\bar{P}R + Q\bar{P}S$

Sol $Q[P + \bar{P}(R + \bar{R}S)]$

$X + YZ$

$Q[(P + \bar{P})(P + R + \bar{R}S)]$

$\rightarrow Q[P + (R + \bar{R})(R + S)]$

$\rightarrow QP + QR + QS //$

Que Simplify

$$(\underbrace{P + \bar{Q} + \bar{R}}_A) (\underbrace{P + \bar{Q} + R}_B) (\underbrace{P + Q + \bar{R}}_C)$$

Sol

$$(P + \bar{Q} + \bar{R}R) (P + Q + \bar{R})$$

{ distributive law

$$\rightarrow (P + \bar{Q}) (P + Q + \bar{R})$$

$$\rightarrow P + P\bar{Q} + P\bar{R} + P\bar{Q}\bar{R} + \bar{Q}\bar{R}$$

$$\rightarrow P(1 + \bar{R}) + P(\bar{Q} + \bar{R}) + \bar{Q}\bar{R}$$

$$\rightarrow P + P + \bar{Q}\bar{R} \Rightarrow P + \bar{Q}\bar{R} //$$

$$\rightarrow P + (Q + \bar{Q})(Q + \bar{R})$$

$$\rightarrow P + Q(Q + \bar{R})$$

$$\rightarrow P + Q + Q\bar{R} //$$

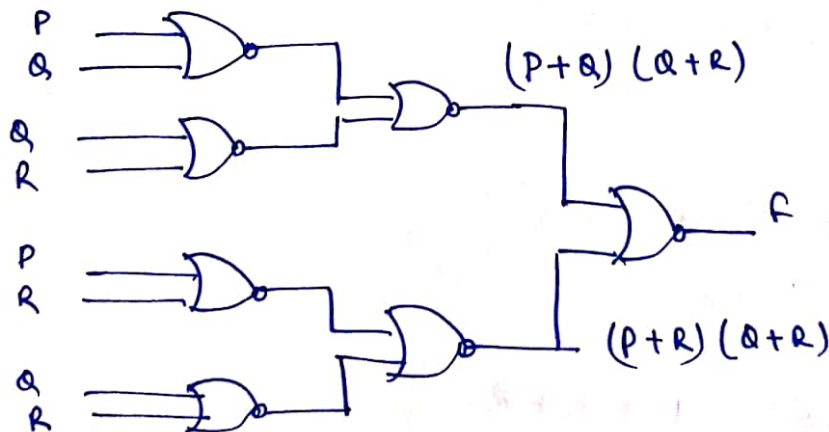
$$\rightarrow P + Q$$

$$P + \bar{Q}(Q + \bar{R})$$

$$P + \bar{Q}Q + \bar{Q}\bar{R}$$

$$\Rightarrow P + \bar{Q}\bar{R}$$

Que. $F = ?$



$$F = (P + Q)(Q + R) + (P + R)(Q + R)$$

$$= (Q + R)(P + Q + R)$$

$$= \overline{Q + R + P \cdot 0} // = \overline{Q + R} //$$

$$\{ (Q + R + 0)(Q + R + P) \\ Q + R + 0 \cdot P$$

* Representation of Boolean Expressions :-

- i) Sum of product form (SOP)
- ii) Product of Sum form (POS)

Product term

$$F = A \bar{B} C$$

$$F = \bar{A} B \bar{C}$$

$$F = A C$$

Sum terms

$$F = \bar{A} + B + C$$

$$F = A + B + \bar{C}$$

$$F = \bar{B} + C$$

Sum of product

$$F = A \bar{B} C + \bar{A} B \bar{C} + A C$$

Product of Sum terms

$$F = (\bar{A} + B + C) (A + B + \bar{C}) (\bar{B} + C)$$

This is not standard form of POS & SOP
In standard form there is no missing variable in all the terms.

Missing

$$F = A \bar{B} C + \bar{A} B \bar{C} + A C$$

$$= A \bar{B} C + \bar{A} B \bar{C} + A C (B + \bar{B})$$

$$= A \bar{B} C + \bar{A} B \bar{C} + A B C + A \bar{B} C$$

$$\Rightarrow A \bar{B} C + \bar{A} B \bar{C} + A B C \quad // \quad \text{now this is in standard SOP form}$$

$$F = (\bar{A} + B + C) (A + B + \bar{C}) (B + C + A \cdot \bar{A})$$

$$= (\bar{A} + B + C) (A + B + \bar{C}) (A + B + C) (\bar{A} + B + C)$$

now this is std.
POS

• All the product terms in standard SOP are referred as minterm.

• All the Sum terms in standard Pos are referred as Maxterm

<u>Minterm table</u>			<u>minterms</u>		<u>Dec</u>
A	B	C			
0	0	0	$\bar{A} \bar{B} \bar{C}$	m_0	0
0	0	1	$\bar{A} \bar{B} C$	m_1	1
0	1	0	$\bar{A} B \bar{C}$	m_2	2
0	1	1	$\bar{A} B C$	m_3	3
1	0	0	$A \bar{B} \bar{C}$	m_4	4
1	0	1	$A \bar{B} C$	m_5	5
1	1	0	$A B \bar{C}$	m_6	6
1	1	1	$A B C$	m_7	7

Note: Normal Variable $\longleftrightarrow 1$ Minterm or Pos
Complemented Var. $\longleftrightarrow 0$

<u>Max term table</u>			<u>Maxterm</u>		
A	B	C			
0	0	0	$A + B + C$	M_0	0
0	0	1	$A + B + \bar{C}$	M_1	1
0	1	0	$A + \bar{B} + C$	M_2	2
0	1	1	$A + \bar{B} + \bar{C}$	M_3	3
1	0	0	$\bar{A} + B + C$	M_4	4
1	0	1	$\bar{A} + B + \bar{C}$	M_5	5
1	1	0	$\bar{A} + \bar{B} + C$	M_6	6
1	1	1	$\bar{A} + \bar{B} + \bar{C}$	M_7	7

Note Maximum or SOP

'1' $\longleftrightarrow$ Complemented Var.
'0' $\longleftrightarrow$ Normal Var.

eg. $F = A\bar{B}C + \bar{A}B\bar{C} + ABC$

$$\Sigma m(101 + 010 + 111)$$

$$= \boxed{\Sigma m(5, 2, 7)} \quad \text{Representation in std. SOP}$$

$$= m_5 + m_2 + m_7$$

eg. $f = (\bar{A} + \bar{B} + C)(A + \bar{B} + C)(A + B + \bar{C})$

$$= \Pi(110, 010, 001)$$

$$= \Pi(6, 2, 1)$$

$$= M_6 \cdot M_2 \cdot M_1 = \boxed{\Pi M(1, 2, 6)} \quad \text{std Pos} //$$

Que. $F = A + \bar{B}C$

i) SOP std:

$$A(B + \bar{B})(C + \bar{C}) + \bar{B}C(A + \bar{A})$$

$$\rightarrow A[BC + B\bar{C} + \bar{B}C + \bar{B}\bar{C}] + A\bar{B}C + \bar{A}\bar{B}C$$

$$\rightarrow ABC + AB\bar{C} + A\bar{B}C + \underline{A\bar{B}C} + A\bar{B}\bar{C} + \bar{A}\bar{B}C$$

$$= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}C + \bar{A}\bar{B}C$$

$$= \Sigma m(7, 6, 4, 5, 1)$$

ii) POS std:

$$f = A + \bar{B}C$$

$$= A + \cancel{A\bar{A}} + \cancel{B\bar{B}} + \bar{B}C + \cancel{A\bar{A}}$$

$$= (\cancel{A} + \cancel{B} + C)(\cancel{A} + \bar{B})$$

first change in POS

$$f = (A + C)(A + \bar{B})$$

$$\begin{aligned}
 F &= (A + \bar{B} + C\bar{C})(A + C + B\bar{B}) \\
 &= (A + \bar{B} + C)(A + \bar{B} + \bar{C})(A + \overbrace{C+B}^1)(A + \overbrace{C+\bar{B}}^1) \\
 &= \prod (010, 011, 000, 010) \\
 &= \prod m(0, 2, 2, 3) \\
 &= M_0 \cdot M_2 \cdot M_2 \cdot M_3 \\
 &= M_0 M_2 M_3 \quad //
 \end{aligned}$$

note:- There is no need to find both Maxterm & Min term of any function. And only Maxterm or Min and find missing term to obtain Min or Maxterm

$$\begin{aligned}
 &\sum m(1, 4, 5, 6, 7) \\
 &\prod M(0, 2, 3)
 \end{aligned}$$

- SOP & POS are NOT complement of Each other
SOP and POS are two different ways to represent any boolean function.

Que. Find Standard SOP form

$$\begin{aligned}
 \textcircled{1} \quad f(A, B, C) &= \bar{A}BC + AB\bar{C} \\
 &\quad \uparrow \quad \uparrow \\
 &\quad \text{MSB} \quad \text{LSB} \\
 &= \sum m(011, 110) \\
 &= \sum m(3, 6)
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{2} \quad f(B, A, C) &= \bar{A}BC + AB\bar{C} \\
 &\quad \uparrow \quad \uparrow \\
 &\quad \text{MSB} \quad \text{LSB} \\
 &= \sum m(101, 110) \\
 &= \sum m(5, 6)
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{3} \quad f(C, A, B) &= \bar{A}BC + AB\bar{C} \\
 &\quad \uparrow \quad \uparrow \\
 &\quad \text{MSB} \quad \text{LSB} \\
 &= \sum m(101, 011) \\
 &= \sum m(5, 3) = \sum m(3, 5)
 \end{aligned}$$

Que. $f = A + \overline{B}C$

A	B	C	
X	0	1	
0	0	1	→ 1
1	0	1	→ 5

A	B	C	
1	X	X	
1	0	0	→ 4
1	0	1	→ 5
1	1	0	→ 6
1	1	1	→ 7

Hence $\Sigma m(1, 4, 5, 7)$

Que. $f(A, B, C) = \overline{A}B + BC$

A	B	C	
X	1	1	
0	1	1	→ 3
1	1	1	→ 7

A	B	C	
0	1	X	
0	1	0	→ 2
0	1	1	→ 3

$\Sigma m(2, 3, 7) //$

Que $f(A, B, C) = \overline{A} + \overline{B}C$

A	B	C	
X	0	1	
0	0	1	→ 1
1	0	1	→ 5

A	B	C	
0	X	X	
0	0	0	→ 0
0	0	1	→ 1
0	1	0	→ 2
0	1	1	→ 3

$\Sigma m(0, 1, 2, 3, 5) //$

Que. $f(C, A, B) = AC + \bar{B}C$

	A	B		C	A	B	
	1	1	X	1	1	0	
7 ←	1	1	1	1	1	0	→ 6
6 ←	1	1	0	1	0	0	→ 4

$\Sigma m(4, 6, 7)$ //

Que. Identify whether SOP or POS

- i) $A + B \rightarrow$ POS
- ii) $AB \rightarrow$ SOP note
- iii) $A \oplus B \rightarrow$ Neither SOP nor POS
- iv) $A \odot B \rightarrow$ Neither SOP nor POS
- v) $\overline{A+B} \rightarrow$ Neither SOP nor POS
- vi) $\overline{A \cdot B} \rightarrow$ Neither SOP nor POS
- vii) $AB + C \rightarrow$ SOP
- viii) $(A+B)(A+C) \rightarrow$ POS

Que. find SOP POS

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

SOP = $\bar{A}B$

POS = $\overline{AB} + \overline{A\bar{A}} + \overline{B\bar{B}} + \bar{A}B$

A	B	Y	
0	0	0	} POS = (A+B)(A+B̄)(Ā+B)
0	1	0	
1	0	0	
1	1	1	} SOP = AB

$$\begin{aligned}
 Y_{POS} &= (A+B)(A+B̄)(Ā+B) \\
 &= (A+B̄B)(Ā+B) \\
 &= A\bar{A} + AB \\
 &= AB
 \end{aligned}$$

$$Y_{POS} = Y_{SOP}$$

Que. find SOP & POS

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

$$POS = (A+B)(Ā+B)$$

$$SOP = \bar{A}\bar{B} + AB$$

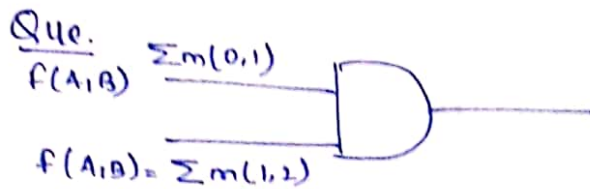
Note:- Reason why '1' Considered for SOP & '0' Considered for POS in truth table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

$$\begin{aligned}
 SOP &= \cancel{\bar{A}\bar{B} \cdot 0} + \cancel{\bar{A}B \cdot 0} + \cancel{A\bar{B} \cdot 0} + AB \cdot 1 \\
 &= AB \text{ i.e. correspond to } 1
 \end{aligned}$$

$$\begin{aligned}
 POS &= (A+B+0)(A+B̄+0) \\
 &\quad (\bar{A}+B+0)(\bar{A}+\bar{B}+1) \quad \text{---}
 \end{aligned}$$

Hence 0 term for POS



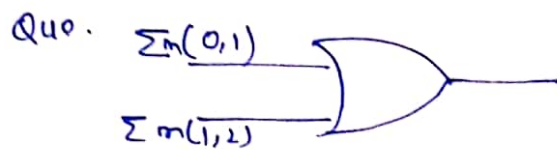
Sol: $(\bar{A}\bar{B} + \bar{A}B)(\bar{A}B + AB)$

{ Distributive law

$\rightarrow \bar{A}B + \bar{A}\bar{B}AB$

$\bar{A}B$

$= \sum m(1) //$

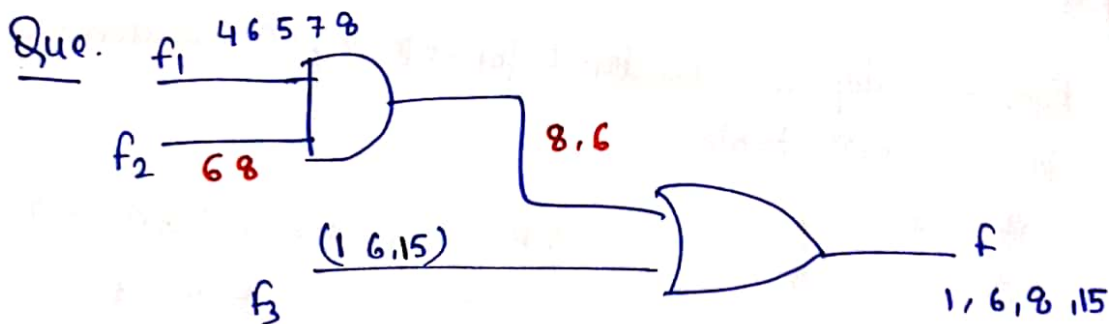


$(\bar{A}\bar{B} + \bar{A}B) + (\bar{A}B + AB)$

$\rightarrow \bar{A}\bar{B} + \bar{A}B + \bar{A}B$

$= \sum m(0, 1, 2)$

Concept:- AND gate pass term which are common in i/p
OR gate pass term which present in all i/p.



$f_1 = \sum m(4, 5, 6, 7, 8)$

$f_2 = \sum m(1, 6, 15)$

$f_2 = ?$

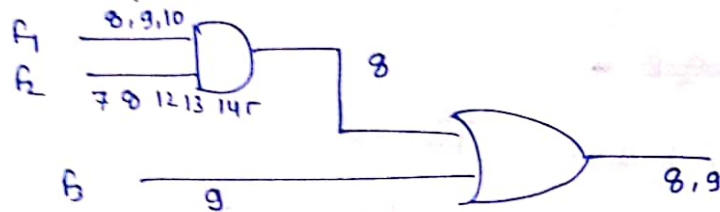
$f = \sum m(1, 6, 8, 15)$

$$11) \quad f_1 = \sum m(8, 9, 10)$$

$$f_2 = \sum m(7, 8, 12, 13, 14, 15)$$

$$f = \sum m(8, 9)$$

$$f_3 = 9$$



$$f_3 = \sum m(9)$$

Que. $f(P, Q, R) = PQ + QR + PR$

$$= PQR + PQ\bar{R} + \bar{P}QR + PQR + PQR + P\bar{Q}R$$

$$= PQR + PQ\bar{R} + \bar{P}QR + P\bar{Q}R$$

$$\quad \quad \quad 7 \quad \quad 6 \quad \quad 3 \quad \quad 5$$

$$= \sum m(3, 5, 6, 7)$$

Que. $f(P, Q, R) = PQ + Q\bar{R} + P\bar{R}$

Sol

PQR	PQR	PQR
1 1 0	0 1 0	1 0 0
1 1 1	1 1 0	1 1 0

$$6, 7, 2, 6, 4, 6$$

$$\Rightarrow \sum m(2, 4, 6, 7)$$

Que. The boolean expression

$f(P, Q, R) = \pi(0, 5)$ is to be implemented using only 2 I/P gates which are these gates.

Sol $\Pi(0, 5)$

$\Rightarrow \cancel{PQR}$

$\Rightarrow (P + Q + R)(\bar{P} + \bar{Q} + \bar{R})$

$\Rightarrow P\bar{P} + Q + (P + R)(\bar{P} + \bar{R})$

$\Rightarrow Q + P\bar{P} + P\bar{R} + R\bar{P} + R\bar{R}$

$\Rightarrow Q + P\bar{R} + R\bar{P}$

$= Q \text{ OR } R \text{ XOR } P //$

$\left\{ \begin{array}{l} \text{Distributive} \\ \text{Law} \end{array} \right.$

* Karnaugh Map (K-Map)

		$\bar{b}$		b	
		0		1	
a	0	0 00	1 01		
	1	2 10	3 11		

a	b	γ
0	0	
0	1	
1	0	
1	1	

		bc	$\overline{b}\overline{c}$	$\overline{b}c$	$b\overline{c}$	
			00	01	11	10
a	0	0 000	1 001	3 011	2 010	
	1	4 100	5 101	7 111	6 110	

a	b	c
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

$ab \backslash c$	$\bar{c}$	c
$\bar{a}\bar{b}$	0 000	1 001
$\bar{a}b$	2 010	3 011
ab	6 110	7 111
$a\bar{b}$	4 100	5 101

$ab \backslash cd$	$\bar{c}\bar{d}$ 00	$\bar{c}d$ 01	$c\bar{d}$ 11	cd 10
$\bar{a}\bar{b}$ 00	0	1	3	2
$\bar{a}b$ 01	4	5	7	6
ab 11	12	13	15	14
$a\bar{b}$ 10	8	9	11	10

$cd \backslash ab$	$\bar{a}\bar{b}$	$\bar{a}b$	$a\bar{b}$	ab
$\bar{c}\bar{d}$	0	4	12	8
$\bar{c}d$	1	5	13	9
$c\bar{d}$	3	7	15	11
cd	2	6	14	10

eg.

- Every term is 1 in Kmap $o/p = 1$
- Diagonal Can't be group together
- group Only $2n^2$ 1's (neighbouring 1's)
- Common term in a group will be o/p of that group

• Try to make group with max^m 1's

eg.

	b	$\bar{b}$	b
a	1	1	
$\bar{a}$			

$$\gamma = \bar{a}$$

	b	$\bar{b}$
a	0	1
$\bar{a}$	1	0

$$\gamma = \bar{a}b + a\bar{b}$$

eg

	bc	$\bar{b}\bar{c}$	$\bar{b}c$	$b\bar{c}$
a	1	1		
$\bar{a}$	1			1

$$\gamma = \bar{c} + \bar{a}b$$

Que1.

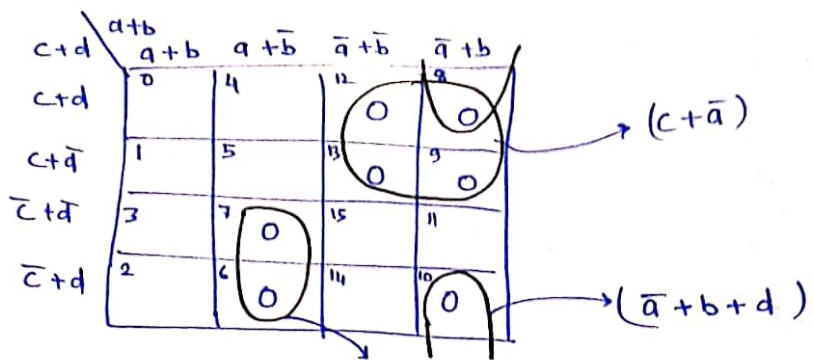
Minimize

	ab	$\bar{a}\bar{b}$	$\bar{a}b$	ab	$a\bar{b}$
$\bar{c}\bar{d}$	1	1			
$\bar{c}d$	1	1			
cd	1			1	1
$c\bar{d}$	1			1	

$$\gamma = \bar{a}\bar{b} + \bar{a}b\bar{c} + cda + \bar{c}dab$$

OR
$$\gamma = \bar{a}\bar{b} + \bar{a}\bar{c} + a\bar{b}cd + abc$$

for above question find Minimal POS form
Sol



$$Y = (c + \bar{a}) + (a + \bar{b} + \bar{c}) + (\bar{a} + b + d)$$

$$Y = (c + \bar{a})(a + \bar{b} + \bar{c}) + (\bar{a} + b + d)$$

note: gray code sequence will be similar

$$Y = \prod M(6, 7, 8, 10, 9, 10, 12, 13)$$

Que. Minimize in pos format

$$f(PQRS) = PR + \bar{P}QS + S$$

$$\sum m = PR(S + \bar{S})(Q + \bar{Q}) + \bar{P}QS(R + \bar{R})$$

P Q R S

1 0 1 0

1 0 1 1

1 1 1 0

1 1 1 1

P Q R S

0 1 0 1

0 1 1 0

P Q R S

0 0 0 1

0 0 1 1

0 1 0 1

0 1 1 1

1 0 0 1

1 0 1 1

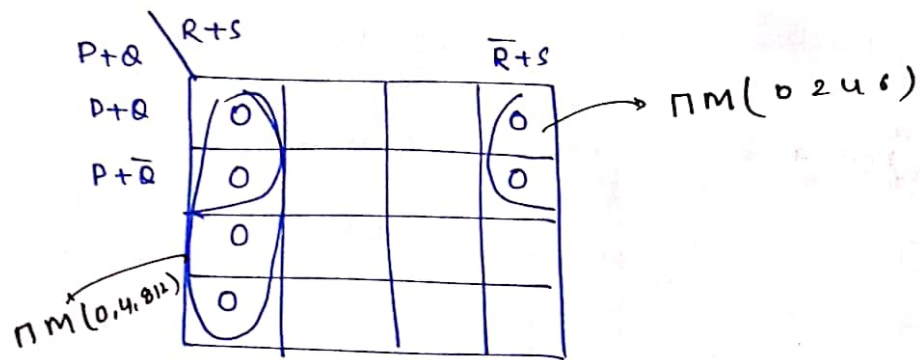
1 1 0 1

1 1 1 1

$$\sum m = \{ 10, 11, 14, 15, 7, 5, 1, 3, 5, 7, 9, 11, 13, 15 \}$$

$$\sum m \{ 1, 3, 5, 7, 9, 10, 11, 13, 14, 15 \}$$

$$\prod M \{ 0, 2, 4, 6, 8, 12 \}$$

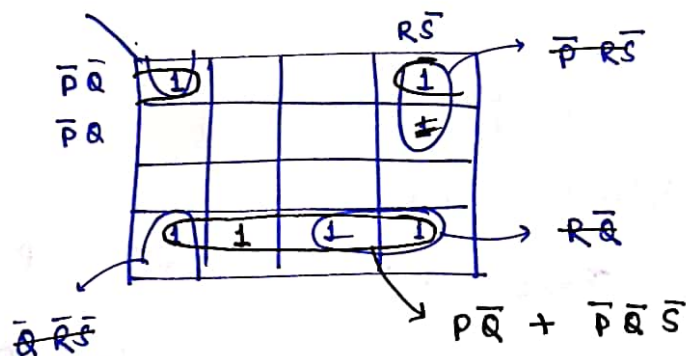


$$Y = (R+S)(P+S) //$$

Q1. Minimal SOP form of Boolean expression

$$Y = \bar{P}\bar{Q}\bar{R}\bar{S} + P\bar{Q}\bar{R}\bar{S} + P\bar{Q}\bar{R}S + P\bar{Q}RS + P\bar{Q}R\bar{S} + P\bar{Q}RS \quad \text{---}$$

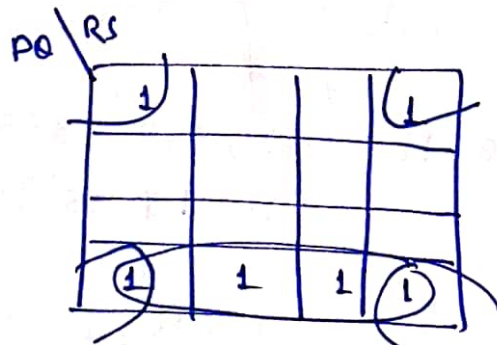
Sol $\Sigma m \{ 0, 2, 4, 6, 10, 11 \}$



$$Y = \bar{P}\bar{R}\bar{S}$$

$$Y = P\bar{Q} + \bar{P}\bar{Q}\bar{S}$$

$$= \bar{Q}(P + \bar{P}\bar{S})$$



$$Y = P\bar{Q} + \bar{Q}\bar{S}$$

Q2.

cd \ ab	1	1		1
	x_1			
	x_2			
	1	1		x_3

find Minimized expression

If $x_3 = 1$ $x_1 = x_2 = 0$

$\bar{c}\bar{d}$	1	1		1
	0			
	0			
	1	1		1

Annotations: $\bar{a}\bar{d}$ (top-left group), $a\bar{b}$ (top-right group), $\bar{b}\bar{d}$ (bottom-right group)

$$y = \bar{b}\bar{d} + \bar{a}\bar{d}$$

If $x_1 = x_2 = 1$ $x_3 = 1$

$$y = \bar{b}\bar{d} + \bar{a}\bar{d} + \bar{a}\bar{b}$$

1	1		1
1			
1			
1	1		1

Annotations: $\bar{a}\bar{b}$ (top-left group), $\bar{b}\bar{d}$ (bottom-right group)

Q3. $f(w, x, y, z) = \sum m(0, 4, 5, 7, 8, 9, 13, 15)$
which of the following is not equivalent to f

Sol

- A) $\bar{x}\bar{y}\bar{z} + \bar{w}x\bar{y} + w\bar{y}z + xz$
- B) $\bar{w}\bar{y}\bar{z} + w\bar{x}\bar{y} + xz$
- C) $\bar{w}\bar{y}\bar{z} + w\bar{x}\bar{y} + x\bar{y}z + x\bar{y}z$
- D) $\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y} + \bar{w}y$

Sol Opt A

Wxyz
1 0 0 0
0 0 0 0

Wxyz
0 1 0 1
0 1 0 0

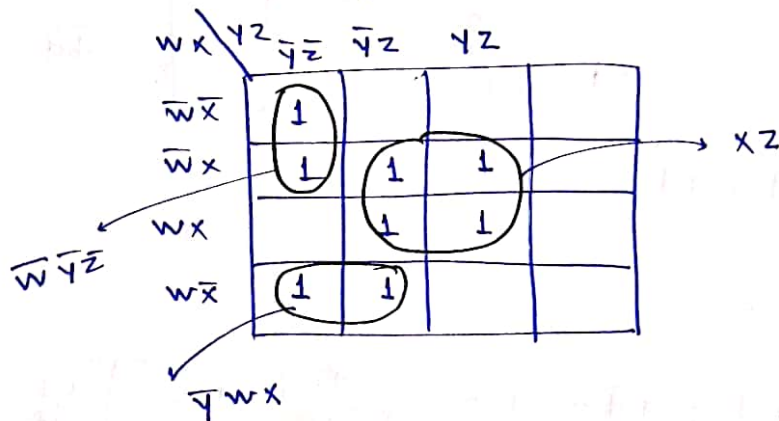
Wxyz
1 0 0 1
1 1 0 1

Wxyz
0 1 0 1
0 1 1 1
1 1 0 1
1 1 1 1

$\Sigma m(0, 4, 5, 8, 9, 7, 13, 15)$
equal to F

Opt B, C, D check in similar way

better Method: using k map



$$F = xz + \bar{w}\bar{y}\bar{z} + \bar{y}wx$$

now opt A: $\bar{x}\bar{y}\bar{z} + \bar{w}x\bar{y} + w\bar{y}z + \textcircled{xz}$

Note
If none of the sop term
present as group in kmap
then that option does not
represent given minterms

Que.

RS \ PQ	PQ	$\bar{P}\bar{Q}$	$\bar{P}Q$	$P\bar{Q}$
$\bar{R}\bar{S}$	0	1	0	0
$\bar{R}S$	0	1	1	1
RS	1	0	1	0
$R\bar{S}$	0	0	1	0

This Quad will not be grouped

$$\bar{R}\bar{P}Q + \bar{R}SP + QS + \bar{P}RS + RPA$$

Wrong

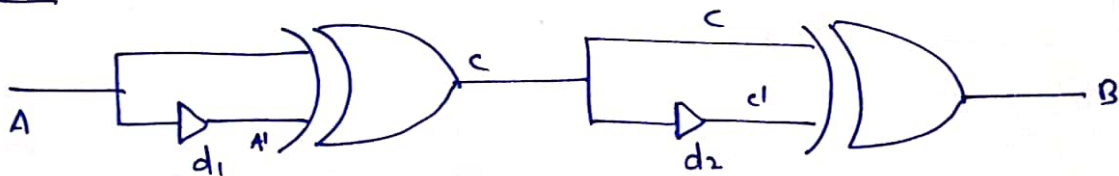
RS \ PQ	PQ	$\bar{P}\bar{Q}$	$\bar{P}Q$	$P\bar{Q}$
$\bar{R}\bar{S}$	0	1	0	0
$\bar{R}S$	0	1	1	1
RS	1	1	1	0
$R\bar{S}$	0	0	1	0

$$\bar{R}\bar{P}Q + \bar{R}SP + \bar{P}RS + RPA$$

Note: Removing.

* Logic Gate Questions

Que.



The non Inverting buffer have delays

$d_1 = 2ns$ & $d_2 = 4ns$ Both XOR gate &

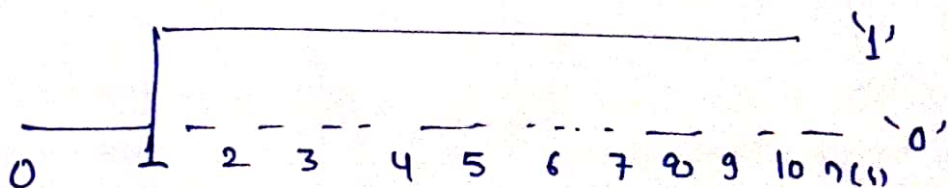
all wire have no delay. Assume all the

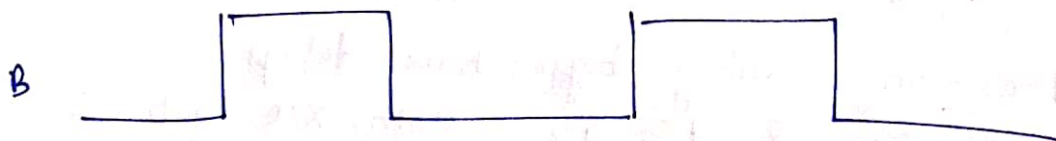
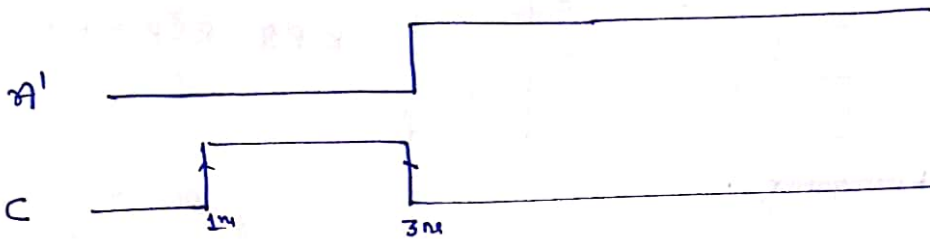
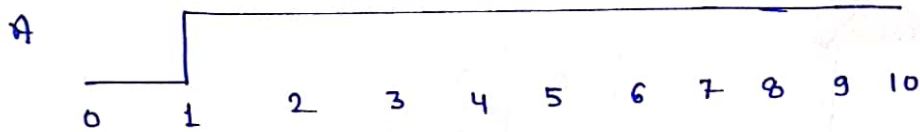
gate i/p and o/p are stable at time

Zero. If following wave form is applied at

i/p A. How many X'sition will be there at o/p B?

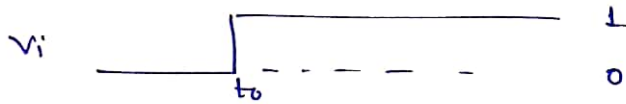
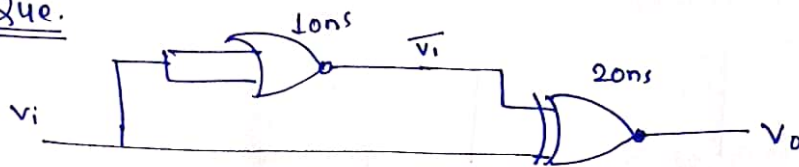
Note
if will not
invert
i/p





Two x's in o/p

Que.



Draw waveform for V_o

note: first Make zero delay waveform and then apply delay

Sol

V_i



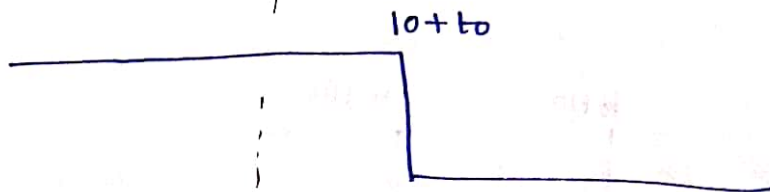
$\overline{V_i}$

Zero delay



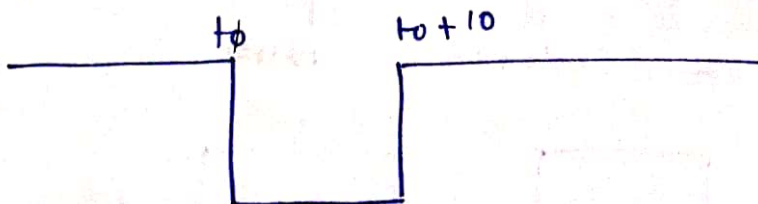
$\overline{V_i}$

10ns



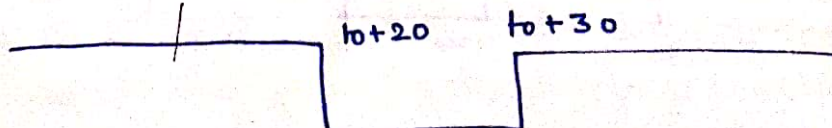
$V_i \oplus \overline{V_i}$

Zero delay

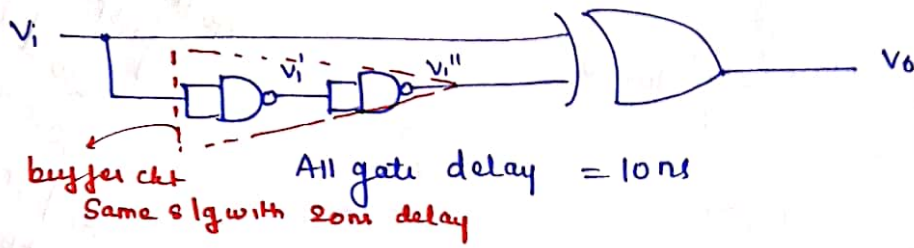
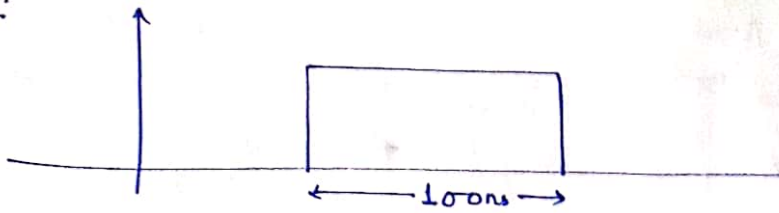


$V_i \oplus \overline{V_i}$

with 20ns delay

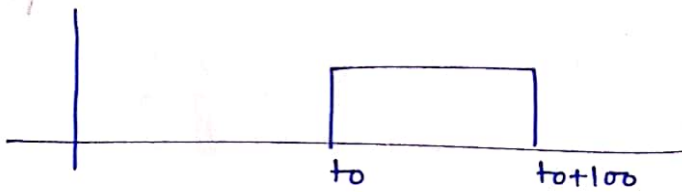


Que.

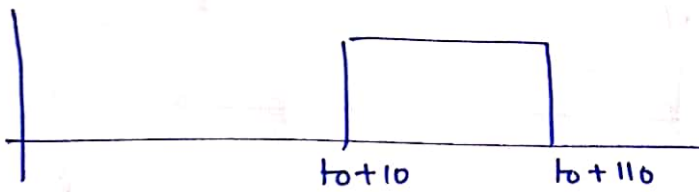


So |

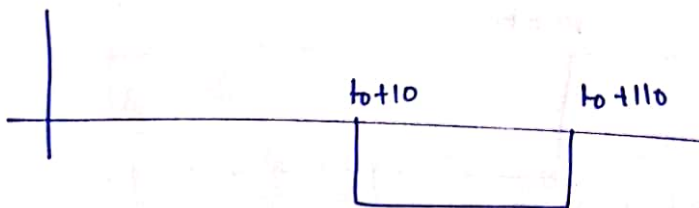
V_i



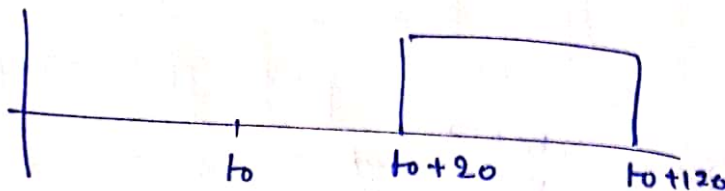
V_i'



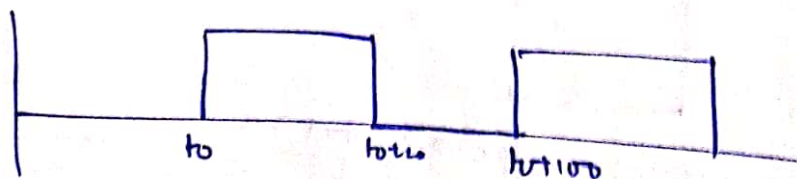
V_i''



V_i''



V_o



//

* Implicants, Prime Implicants & Essential Implicants ÷

1) Implicants

$$\text{Let, } f = \sum m(0, 2, 3, 6, 8, 9, 12, 13, 15)$$

$$i) \quad g_1 = \sum m(0, 3, 8, 12, 15)$$

f is Covering function $g_1 \Rightarrow g_1$ is Subfunction of f i.e. g_1 is an Implicant of f .

$$ii) \quad g_2 = \sum m(2, 6, 9, 13, 15)$$

g_2 is Subfunction of f i.e. g_2 is an Implicant of f .

$$iii) \quad g_3 = \sum m(0, 3, 6, 9, 10^*)$$

f is not Covering function $g_3 \Rightarrow g_3$ is not Subfunction of f
i.e. g_3 is not an Implicant of f .

$$iv) \quad g_4 = \sum m(2, 6, 9, 14^*)$$

f is not Covering function $g_4 \Rightarrow g_4$ is not Subfunction of f
of f i.e. g_4 is not an Implicant of f .

Que 1. $F(A, B, C) = \sum m(1, 5, 7, 8, 10, 12, 13, 15)$

$$h_1 = B\bar{C}$$

$$h_2 = \bar{A}\bar{C}$$

$$h_3 = A\bar{B}\bar{C}$$

$$h_4 = A\bar{C}\bar{B}$$

$$h_5 = AB\bar{C}$$

$$h_6 = B\bar{C}C$$

$$h_7 = \bar{A}B\bar{C}$$

$$h_8 = B\bar{C}C$$

$$h_9 = \bar{A}\bar{B}\bar{C}$$

Sol:- $h_1 = B D$

A	B	C	D	
0	1	0	1	→ 5
0	1	1	1	→ 7
1	1	0	1	→ 13
1	1	1	1	→ 15

$h_2 = \bar{A} \bar{C} D$

A	B	C	D	
0	0	0	1	→ 1
0	1	0	1	→ 5

$h_3 = A \bar{B} \bar{D}$

1	0	0	0	→ 8
1	0	1	0	→ 10

$h_4 = A \bar{C} \bar{D}$

A	B	C	D	
1	0	0	0	→ 8
1	1	0	0	→ 12

$h_5 = A B \bar{C} D$

1	1	0	0	→ 12
1	1	0	1	→ 13

$h_6 = A B C D$

0	1	0	1	→ 5
1	1	0	1	→ 13

$h_7 = \bar{A} B D$

A	B	C	D	
0	1	0	1	→ 5
0	1	1	1	→ 7

$h_8 = B \bar{C} \bar{D}$

A	B	C	D	
0	1	0	0	→ 4*
1	1	0	0	→ 12

It is not Implicant

h_9

A	B	C	D	
0	0	0	0	→ 0*
0	0	0	1	→ 1

h_9 is not Implicant

Hence only h_8, h_9 are not Implicant of F.

k map

Implicant :- If group possible?

Yes → Implicant

No → not prime Implicant

Prime Implicant :- Are best group possible?

Yes → Prime Implicant

k map for

	00	01	11	10
11	0	1	3	2
01	4	5	7	6
10	12	13	15	14
00	8	9	11	10

Prime Implicants $\rightarrow (5, 7, 13, 15)$

$h_2 (1, 5)$ PI

$h_3 (8, 10)$ PI

$h_4 (8, 12)$ PI

$h_5 (12, 13)$ PI

$h_6 (5, 13)$ NOT prime implicant, They can include in Quad

$h_7 (5, 7)$ NOT prime implicant

$h_1 (5, 7, 13, 15)$ Prime Implicant

If any term cannot come under Quad they are PI

Essential prime Implicants

$h_1 (1, 5)$ EPI due to 1

$h_3 (8, 10)$ EPI due to 10 because 10 has no option of group (8, 10) is compulsory

$h_4 (8, 12)$ NOT EPI (8-10) (12-13) can be possible

$h_5 (12, 13)$ NOT EPI

$h_6 (5, 13)$ NOT EPI

$h_1 (5, 7, 13, 15)$ ~~NOT~~ EPI

Pair is not alternative of Quad Quad itself is best group

due to 7 & 15

Note: Implicant $\rightarrow$ any group

Prime Imp. $\rightarrow$ all possible group, can be other option group

EPI $\rightarrow$ no other option for group

Redundant PI $\rightarrow$ group which can't be formed but exist

Que.

AB \ C0	00	01	11	10
00	1	0	1	1
01	0	1	0	0
11	0	1	1	0
10	1	0	0	1

no. of EPI = 9

Sol (3, 2) (12, 15) (5, 13) (0, 2, 8, 10)

4 essential prime implicants.

Que.

AB \ C0	00	01	11	10
00	1	1	0	1
01	0	0	0	1
11	1	0	0	0
10	1	0	0	1

no. of essential prime imp
= 4

Que. which are the EPI of following Boolean function

$$f(a, b, c) = a'c + ac' + b'c$$

a) $a'c$ & ac' b) $a'c$ & $b'c$ c) ac' only d) ac' & bc'

Sol $\Sigma m(3, 1, 4, 6, 5)$

a \ bc	00	01	11	10
0		1	1	D
1	1	1	1	D

Wrong

a \ bc	00	01	11	10
0		1	1	
1	1	1		1

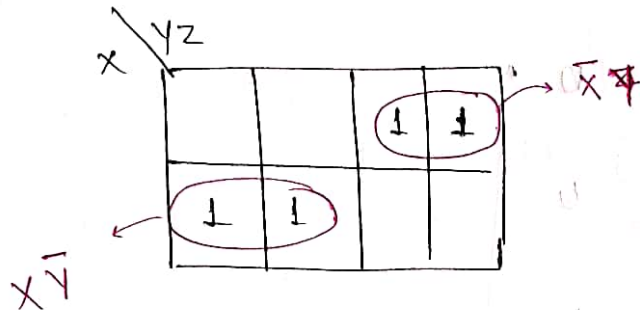
Hence opt a

Que. In SOP function

$$f(x,y,z) = \sum m(2,3,4,5)$$

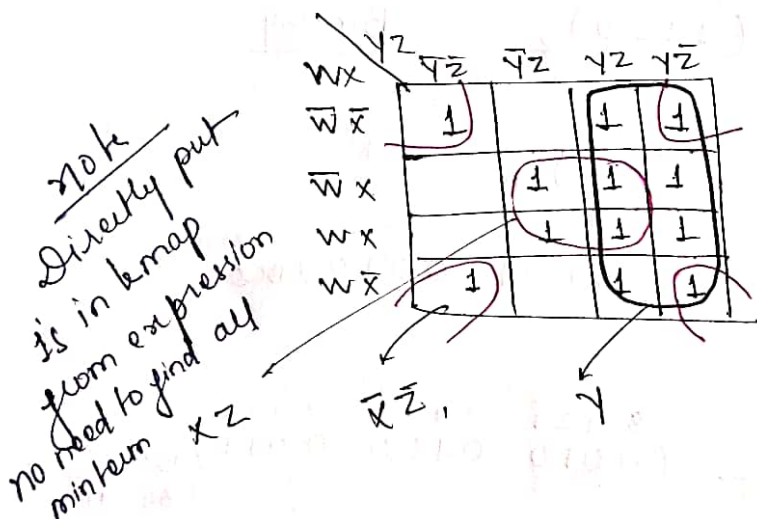
the prime implicants are

- i) $\bar{x}y, x\bar{y}$ ii) $\bar{x}y, x\bar{y}z, x\bar{y}z$ iii) $\bar{x}y\bar{z}, \bar{x}yzy$
 iv) $\bar{x}y\bar{z}, \bar{x}yzy$
 $x\bar{y}z, x\bar{y}z$



Que. $f(w,x,y,z) = wy + xy + \bar{w}xyz + \bar{w}\bar{y} + xz + \bar{x}\bar{y}\bar{z}$

which of the following is complete set of EPI?



$$wy = \begin{matrix} wx & yz \\ 1 & 0 & 1 & 0 & \rightarrow & 10 \\ 1 & 0 & 1 & 1 & \rightarrow & 11 \\ 1 & 1 & 1 & 0 & \rightarrow & 14 \\ 1 & 1 & 1 & 1 & \rightarrow & 15 \end{matrix}$$

$$xy = \begin{matrix} wx & yz \\ 0 & 1 & 1 & 0 & \rightarrow & 6 \\ 0 & 1 & 1 & 1 & \rightarrow & 7 \\ 1 & 1 & 1 & 0 & \rightarrow & 14 \\ 1 & 1 & 1 & 1 & \rightarrow & 15 \end{matrix}$$

$$\bar{w}xyz \rightarrow 7$$

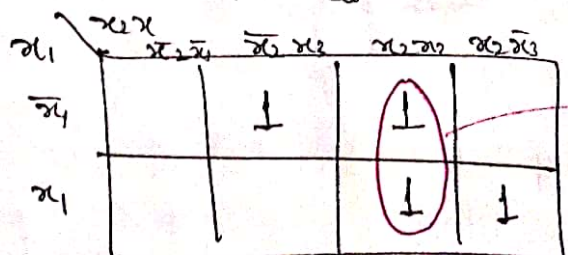
$$\bar{w}\bar{x}y$$

Hence. $xz, \bar{x}\bar{z}, y$ //

(Avoid this calⁿ)

Que $f(x_1, x_2, x_3) = \sum m(1,3,6,7)$

Redundant PI is -



Redundant PI

$x_2 x_3$ i.e.
 Pairs which
 have optⁿ
 to for alternative pairs

* BCD codes

Binary coded decimal :- Every decimal digit is encoded with four binary bit

eg. $(12)_{10} \xrightarrow{\text{Binary}} ()_2$

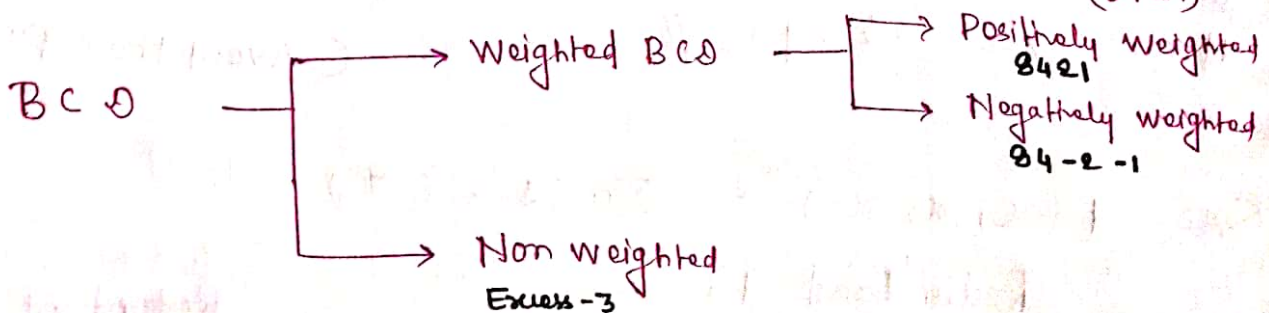
$\frac{12}{2}$	0
$\frac{6}{2}$	0
$\frac{3}{2}$	1
$\frac{1}{2}$	1

$(12)_{10} \longrightarrow (1100)_2$ Binary

eg1 $(12)_{10} \longrightarrow ()_{BCD}$

1 2 $(0001\ 0010)_{BCD}$
0001 0010

eg 2) $(243)_{10} \longrightarrow (0010\ 0100\ 0011)_{BCD}$
(8421)



	8421	84-2-1	XS-3
7	0111	1001	1010
5	0101	1011	1000

$$(57)_{10} = (0101 \ 0111)_{8421} = (1011 \ 1001)_{84-2-1}$$

$$= (1000 \ 1010)_{XS-3}$$

Decimal	(Downed Code & unused) 8421 BCD	(Downed & unused) 84-2-1 BCD	(Downed & unused) XS-3 BCD	(All 16 codes are used) 2421	2421
0	0000	0000	0011	0000	0000
1	0001	0111	0100	0001	0001
2	0010	0110	0101	1 000 0010	0010
3	0011	0101	0110	0011	1001
4	0100	0100	0111	0100	1010
5	0101	1011	1000	0101	0101
6	0110	1010	1001	0110	0110
7	0111	1001	1010	0111	1101
8	1000	1000	1011	1110	00 1110
9	1001	1111	1100	1111	1111

Note: Sequential Code :- By adding 1 to a codeword, we get next codeword
eg. BCD & XS-3

non Sequential Code

eg. 84-2-1 BCD

Advantage of Sequential Codes

→ Arithmetic operation are possible in Sq. Code

eg. 8421 BCD arithmetic

X5-3 arithmetic

* Self Complementing Code

Number System

$(R-1)$'s
Complement

R 's
Complement

eg.

Decimal
($R=10$)

Octal
($R=8$)

Binary
($R=2$)

9's

10's

7's

8's

1's

2's

$$782 \rightarrow \begin{array}{r} 999 \\ 782 \\ \hline 217 \end{array} \quad \begin{array}{r} 217 \\ 1 \\ \hline 218 \end{array}$$

$$\begin{array}{r} 777 \\ 713 \\ \hline 064 \end{array} \quad \begin{array}{r} 064 \\ 1 \\ \hline 065 \end{array}$$

$$\begin{array}{r} 111 \\ 010 \\ \hline 101 \end{array} \quad \begin{array}{r} 101 \\ 1 \\ \hline 110 \end{array}$$

$$x's \text{ comp} = (R-1)'s + 1$$

9's Complement pair

$$9 \longleftrightarrow 0$$

$$8 \longleftrightarrow 1$$

$$7 \longleftrightarrow 2$$

$$6 \longleftrightarrow 3$$

$$5 \longleftrightarrow 4$$

eg. ~~8421~~, 84-2-1, X5-3
Decimal

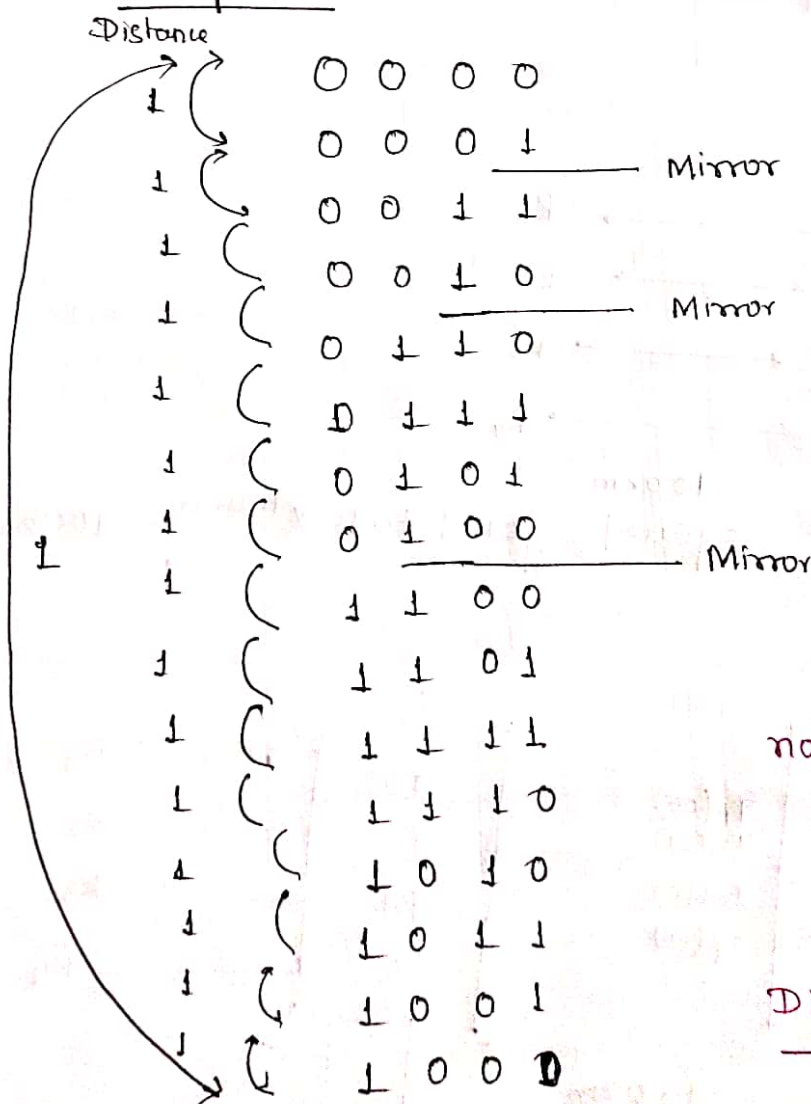
all are having
9's pair

8421 is not Self Complementing

- Self Complementing Code is a Code in which we can obtain Codeword for N 's Complement of N by from the Code of N by Complementing each bit in Code of N

- 2421 is a Self Complementing Code. Any Code which has repeated weight (2 in this case) are Self Complementing.
- Self Complementing property must hold to write Codewords. Hence there are many Sequence of 2421 BCD are possible

* Gray Code



- Gray code is non weighted Code.
- Gray Code is also called Reflective Code
- Gray code is not a BCD Code because Gray Code can be of 2 bit, 3 bit 4 bit 8 bit but BCD is only 4 bit.

note:- Gray Code are Unit distance Code

Distance between Code
→ difference in no. of bit

- Gray Code is also called Cycle Code